Centralized Egress Firewall on AWS

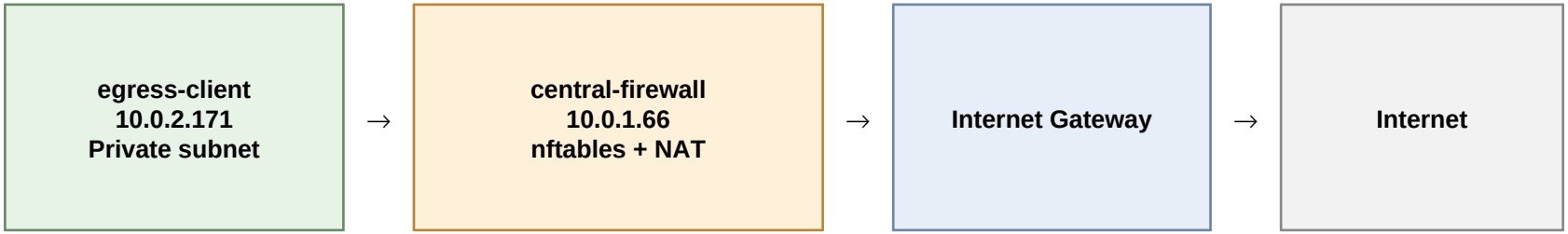
Implementation and Screenshot Evidence Report

A cost-conscious proof of concept using an Amazon EC2 instance with Linux nftables as a centralized outbound firewall/router for a private EC2 client.

Region	Europe (Stockholm) - eu-north-1
VPC	egress-filtering-vpc - 10.0.0.0/16
Firewall	central-firewall - 10.0.1.66
Private client	egress-client - 10.0.2.171
Filtering	nftables FORWARD policy drop; HTTP/HTTPS permitted
NAT	Masquerade for 10.0.2.0/24 through firewall interface ens5

1. Architecture and Traffic Flow

The private client has no public IPv4 address. Internet-bound traffic is directed by the client subnet route table to the firewall EC2 network interface. The firewall forwards only permitted web traffic, performs source NAT, and sends the traffic through the Internet Gateway.



Forwarding policy: established/related traffic is accepted; new TCP traffic from 10.0.2.0/24 is permitted only for destination ports 80 and 443; other forwarded traffic falls through to the default drop policy.

Important router requirement: source/destination checking on the firewall ENI must be disabled, because the firewall forwards packets whose source or destination is another host.

2. Final Validation Summary

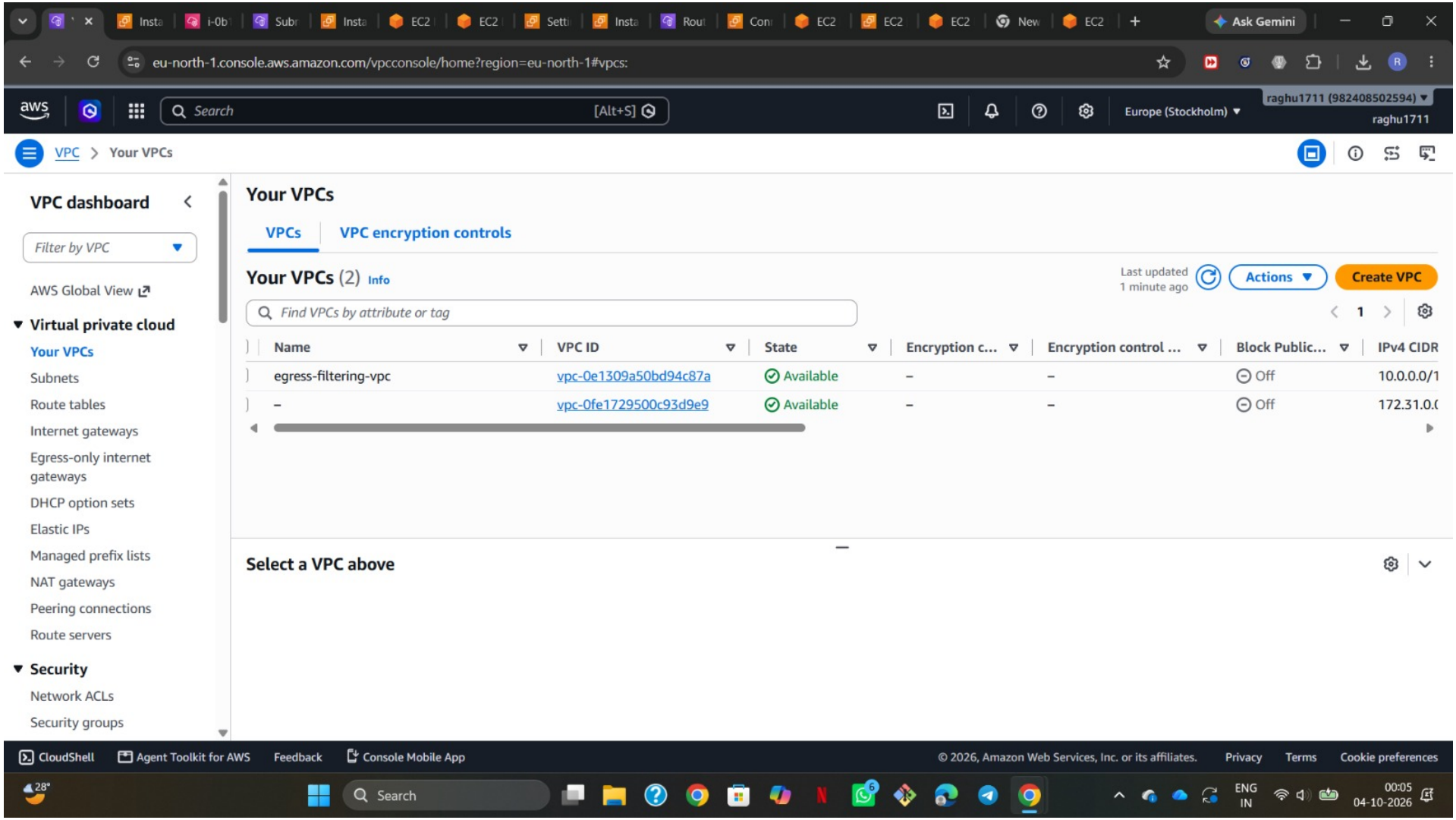
Validation item	Evidence / result
Private client routing	Client uses 10.0.2.1 as its subnet gateway; the AWS subnet route table sends 0.0.0.0/0 to the firewall ENI.
Firewall forwarding	Linux forwarding and nftables are configured on central-firewall.
Stateful filtering	FORWARD chain uses policy drop and accepts established/related return traffic.
Allowed outbound web traffic	TCP destination ports 80 and 443 are accepted for source subnet 10.0.2.0/24.
Source NAT	10.0.2.0/24 is masqueraded on interface ens5.
Internet access	Client receives successful HTTP responses from example.com.
Private management	Systems Manager shows egress-client Online and Session Manager connects successfully.
Rule persistence preparation	/etc/nftables/main.nft contains the clean final rules and passes nft -c syntax validation.

3. Screenshot Evidence

The following pages preserve the full-resolution screenshots in chronological order. Each figure includes a short explanation of what the screenshot demonstrates.

Figure 1. VPC Configuration

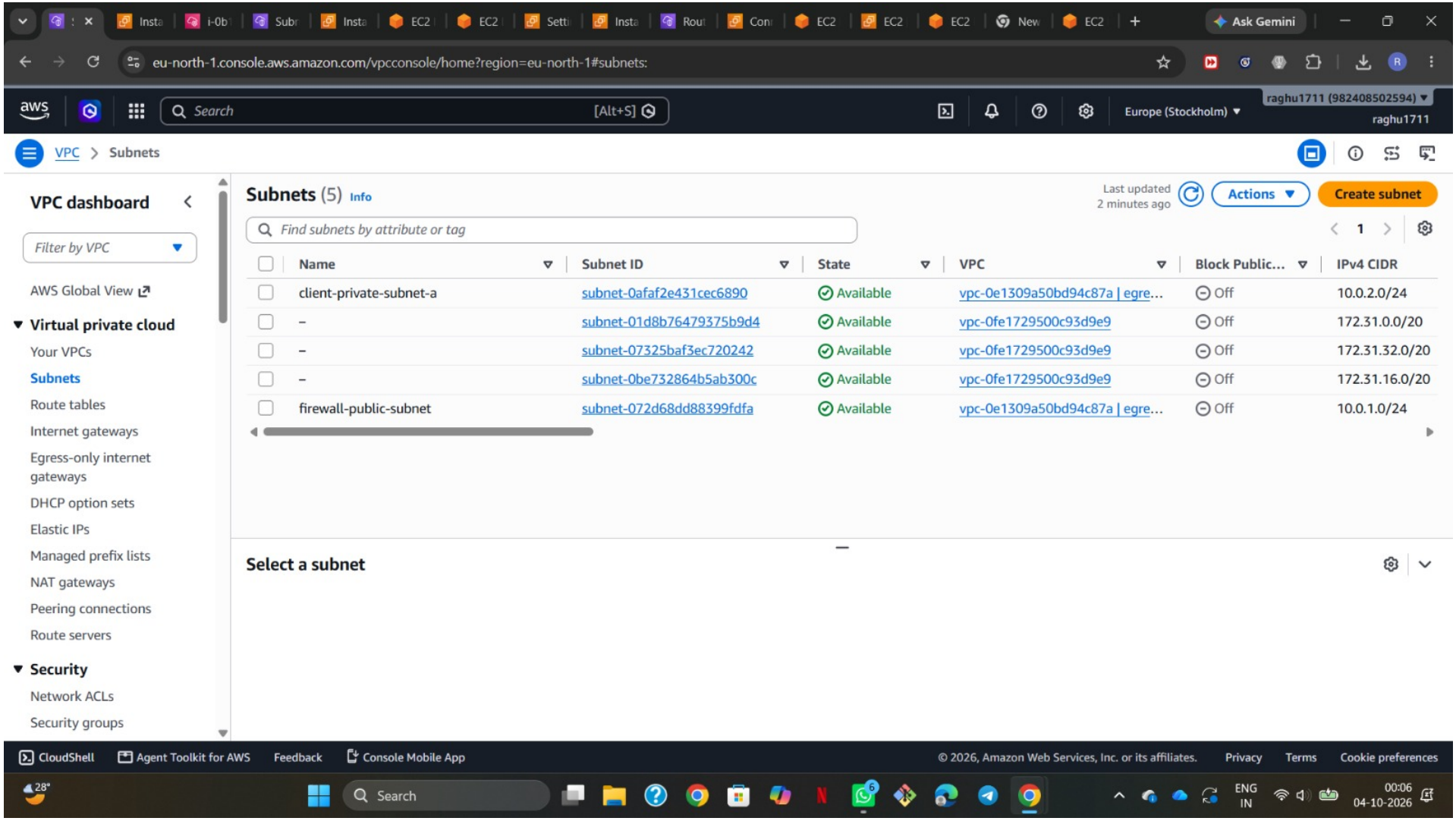
Shows the project VPC used for the centralized egress filtering lab.



Source screenshot: WhatsApp Image 2026-10-04 at 12.06.09 AM.jpeg

Figure 2. Subnet Configuration

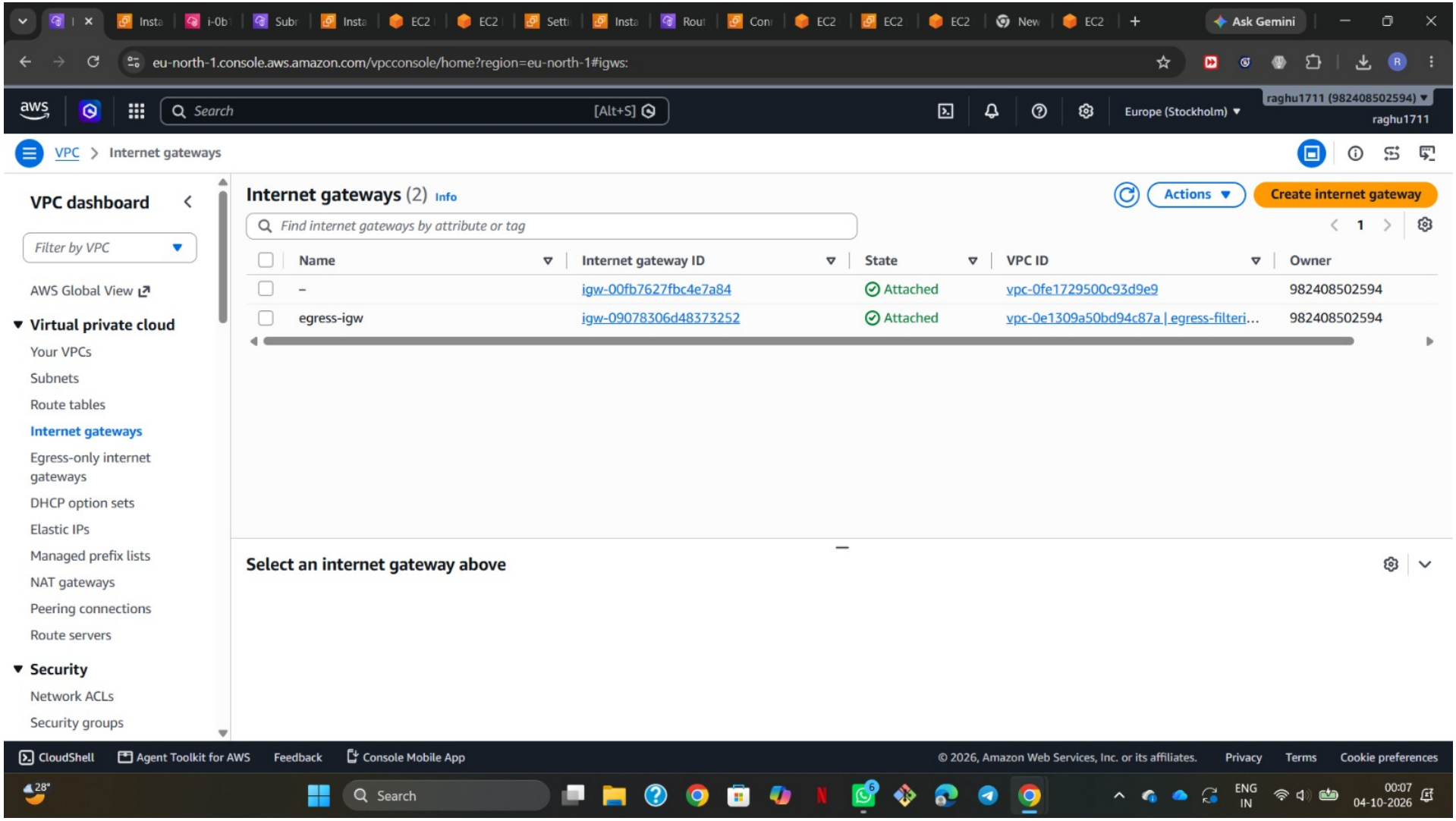
Shows the firewall public subnet and client private subnet created inside the project VPC.



Source screenshot: WhatsApp Image 2026-10-04 at 12.06.41 AM.jpeg

Figure 3. Internet Gateway

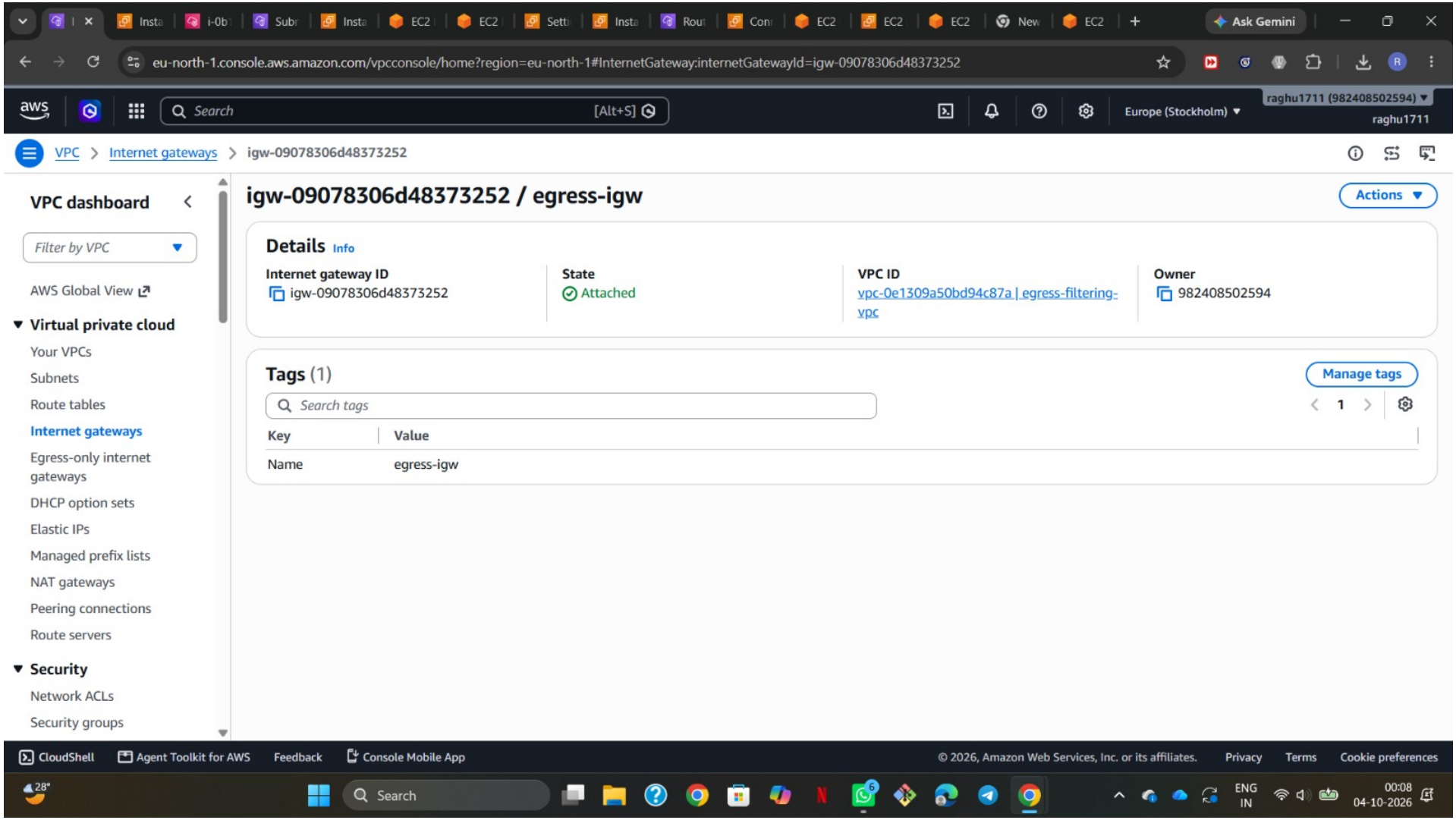
Shows the Internet Gateway attached to the VPC, providing the Internet path for the firewall instance.



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Figure 4. Internet Gateway Details

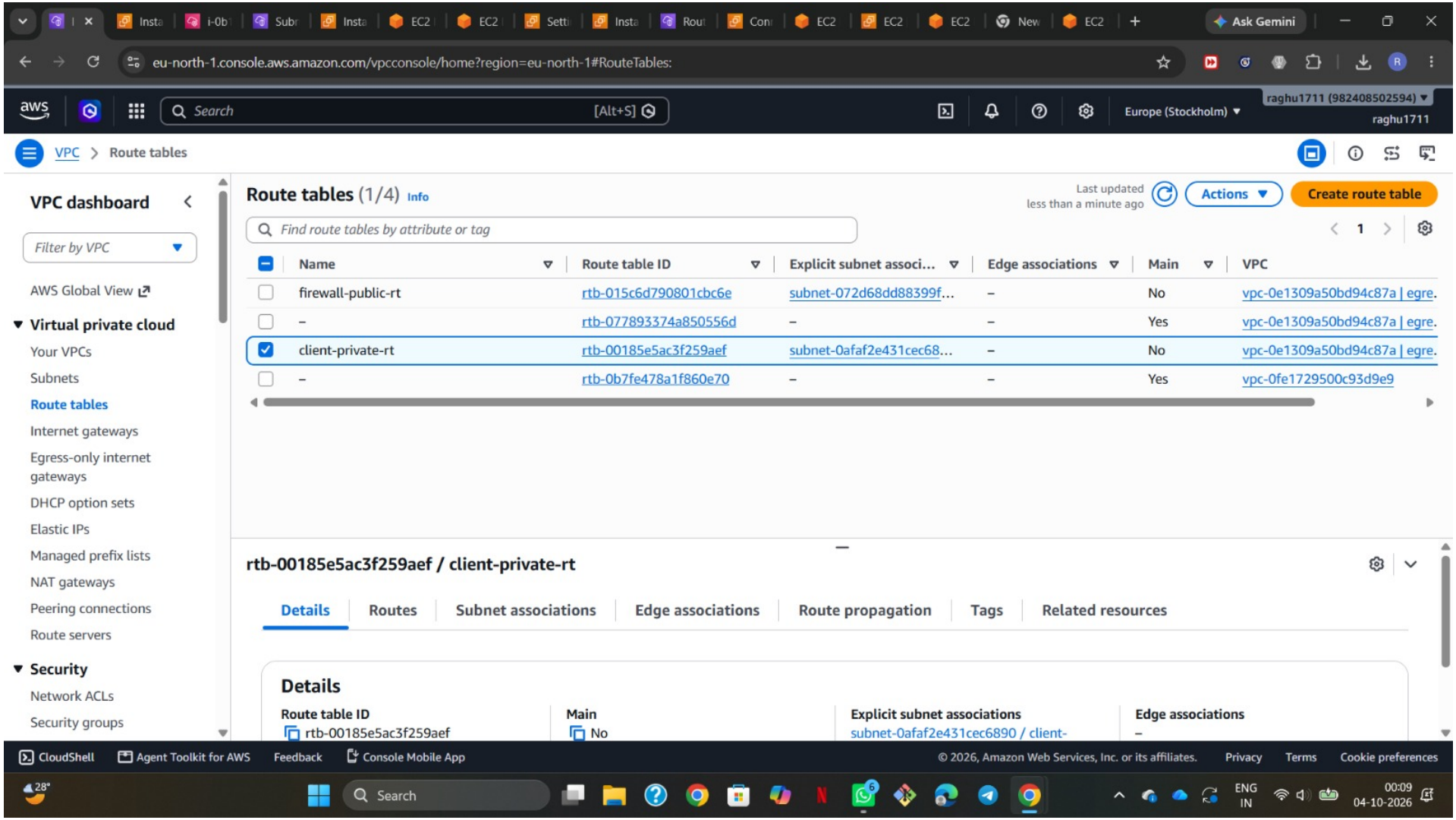
Confirms the selected Internet Gateway and its VPC attachment.



Source screenshot: WhatsApp Image 2026-10-04 at 12.08.38 AM.jpeg

Figure 5. Route Tables Overview

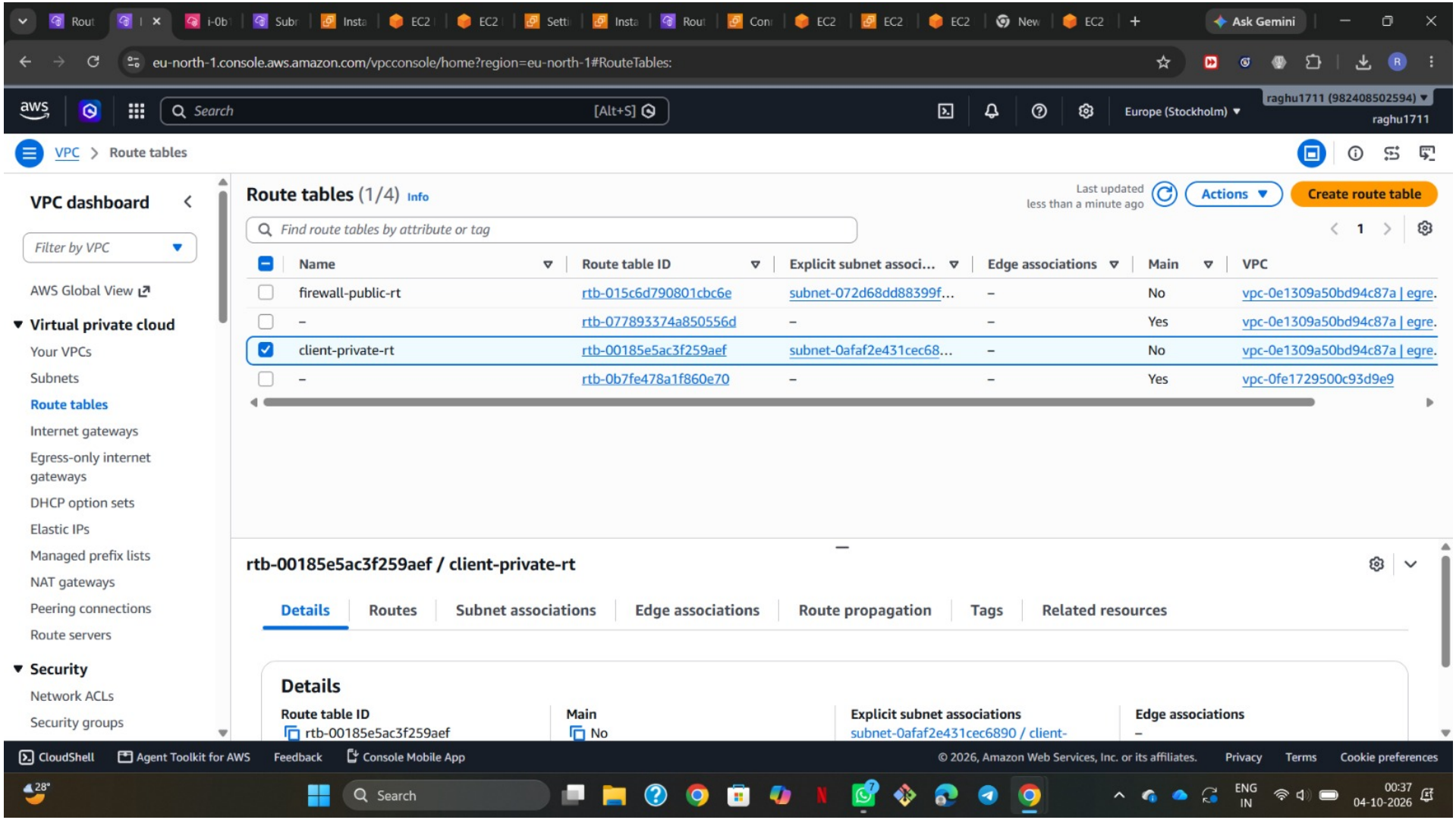
Shows the route tables used to separate the public firewall path from the private client path.



Source screenshot: WhatsApp Image 2026-10-04 at 12.10.06 AM.jpeg

Figure 6. Route Table Selection

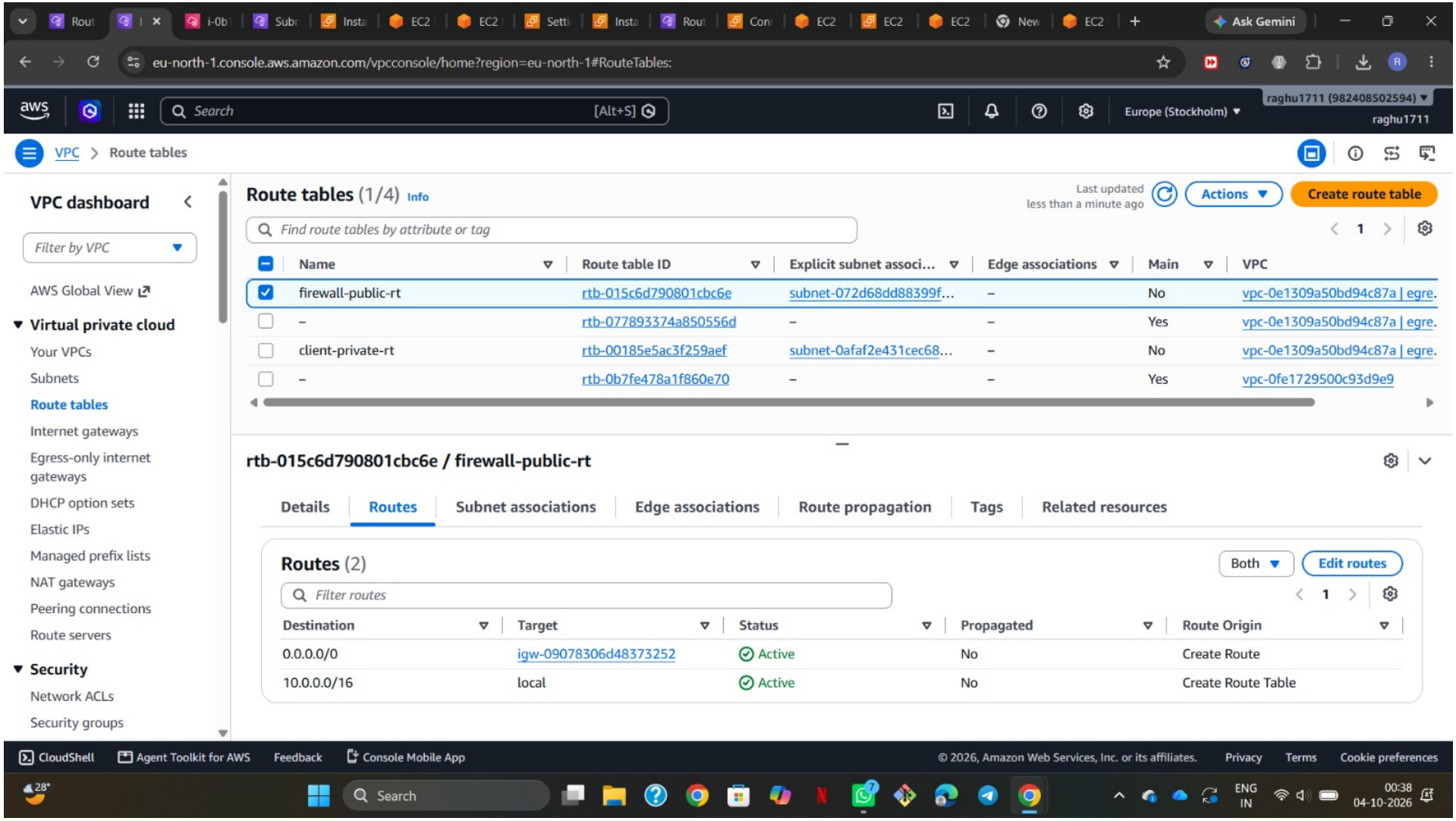
Shows the project route tables and their subnet associations.



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Figure 7. Client Private Route Table

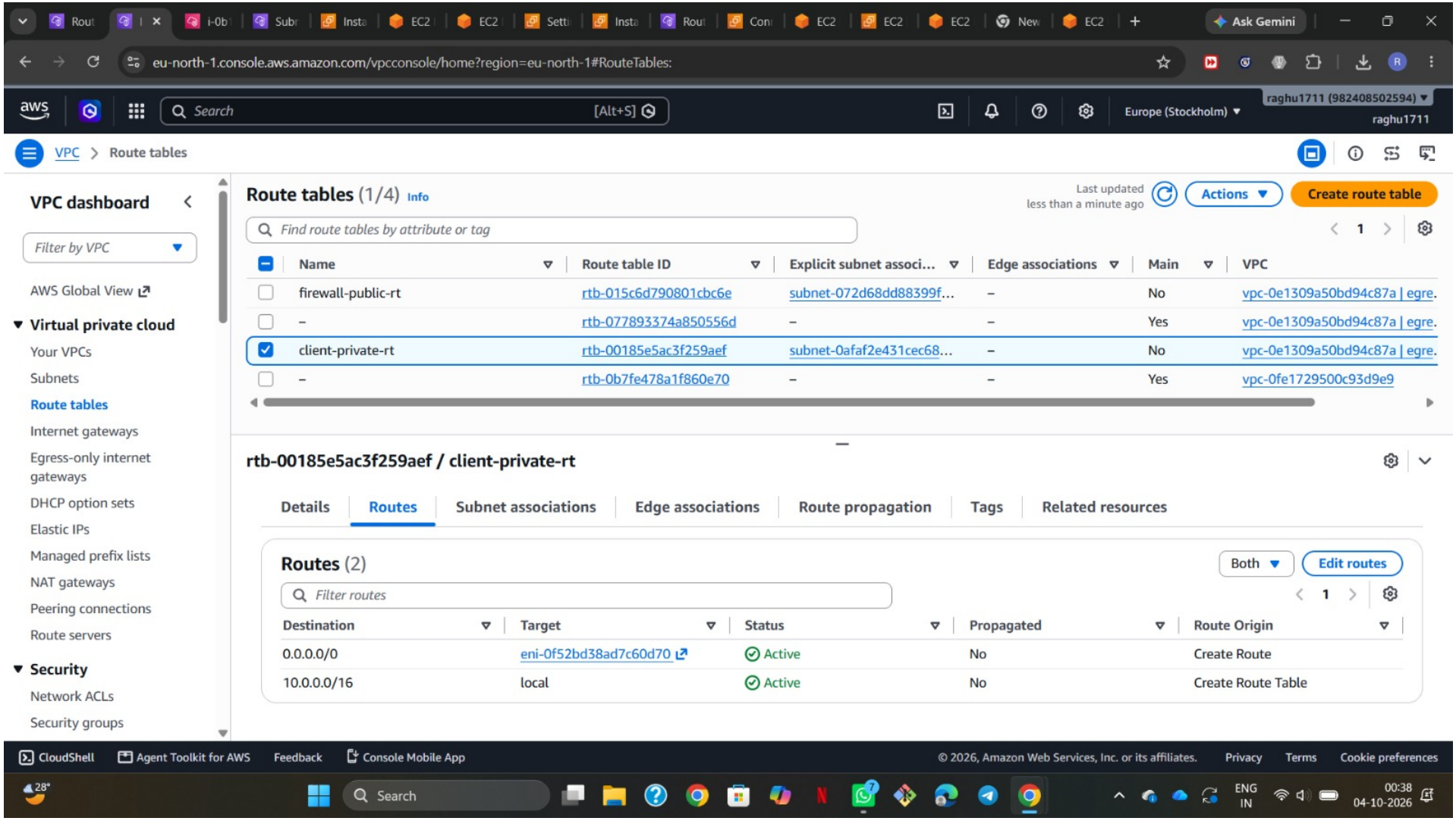
Evidence that the private client subnet uses the firewall network interface as its default IPv4 route.



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Figure 8. Firewall Public Route Table

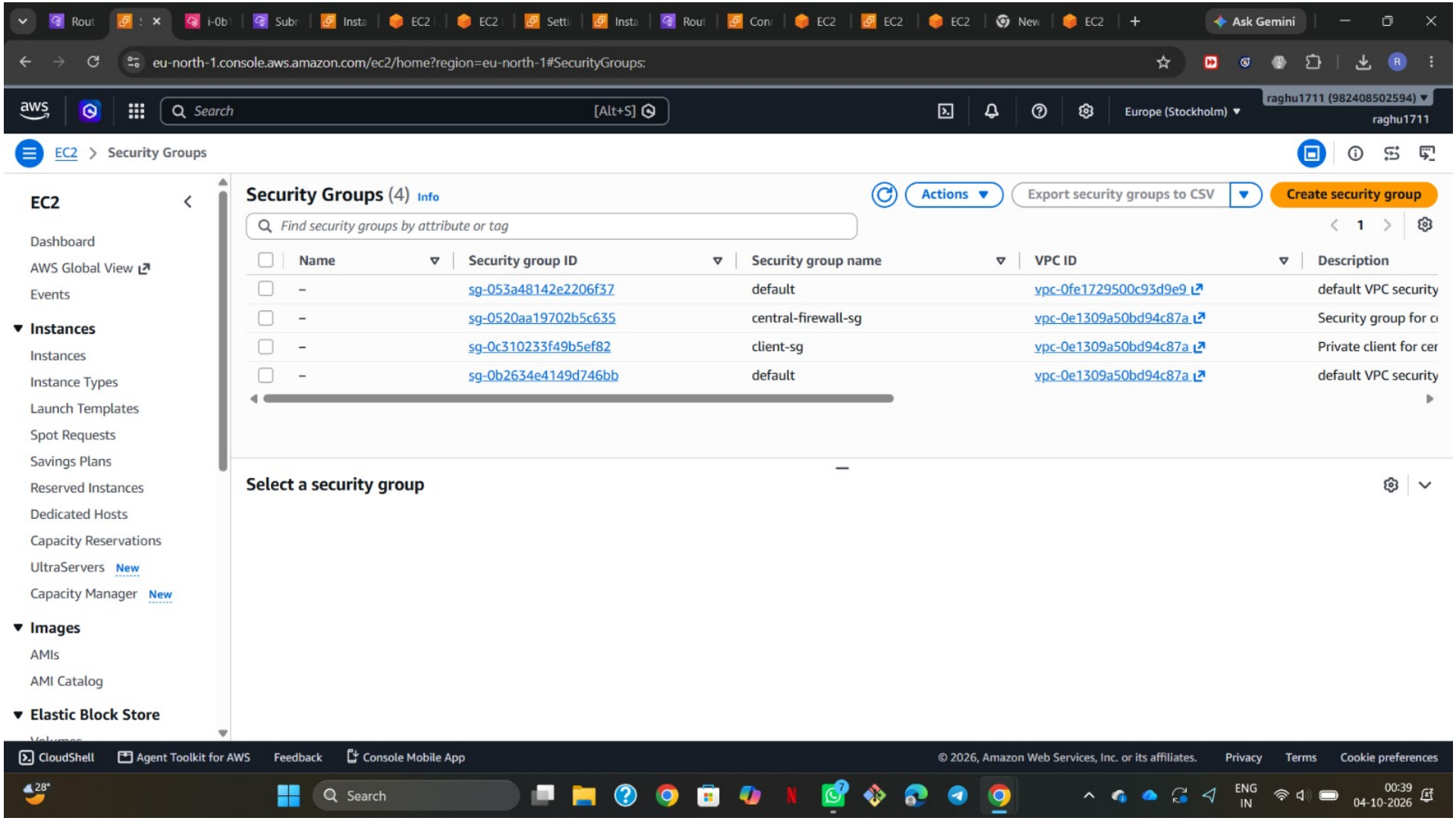
Shows the firewall subnet route table with Internet-bound traffic directed to the Internet Gateway.



Source screenshot: WhatsApp Image 2026-10-04 at 12.38.40 AM.jpeg

Figure 9. Security Groups Overview

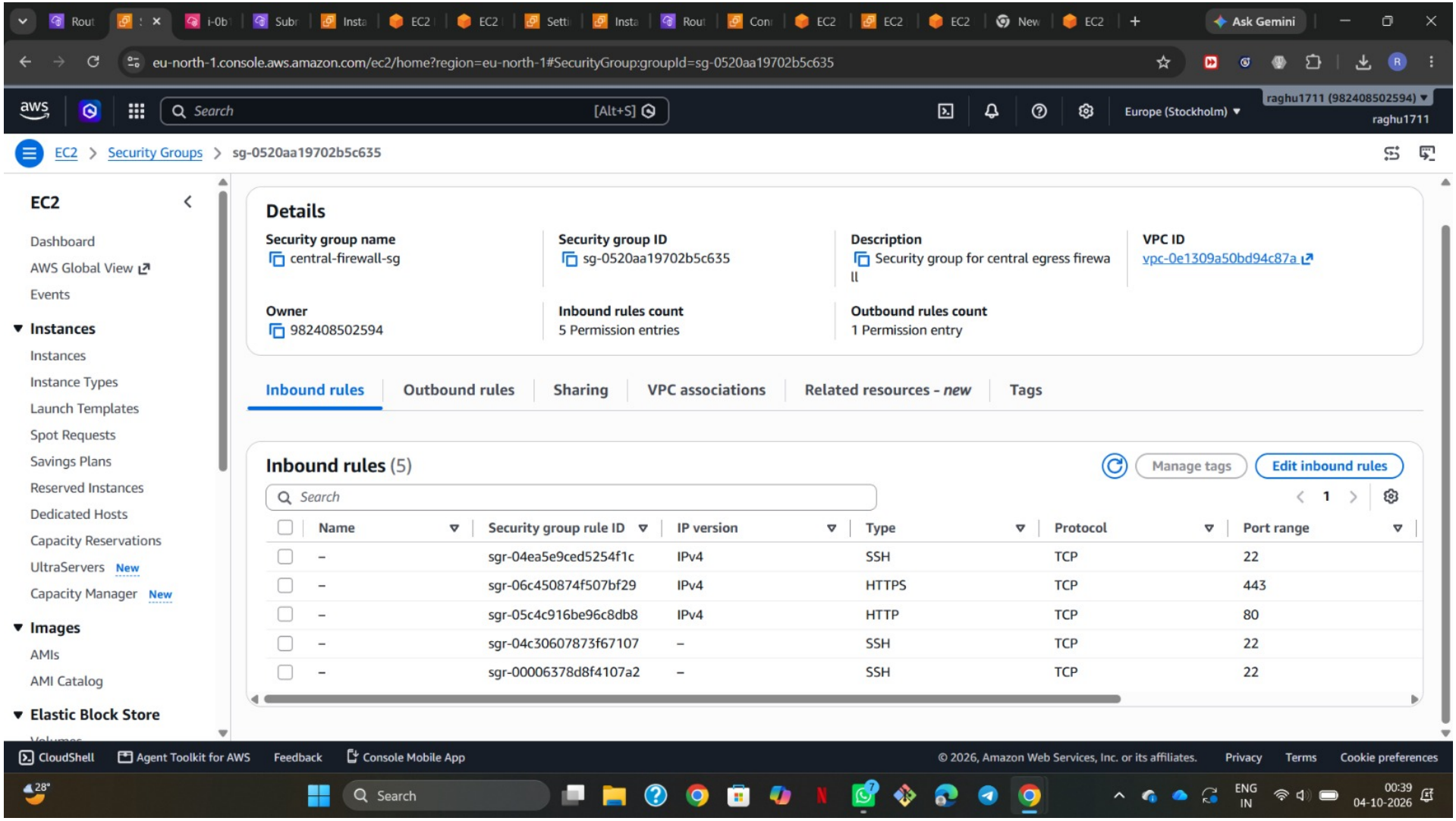
Shows the security groups used to control access to the firewall and client instances.



Source screenshot: WhatsApp Image 2026-10-04 at 12.39.35 AM.jpeg

Figure 10. Client Security Group

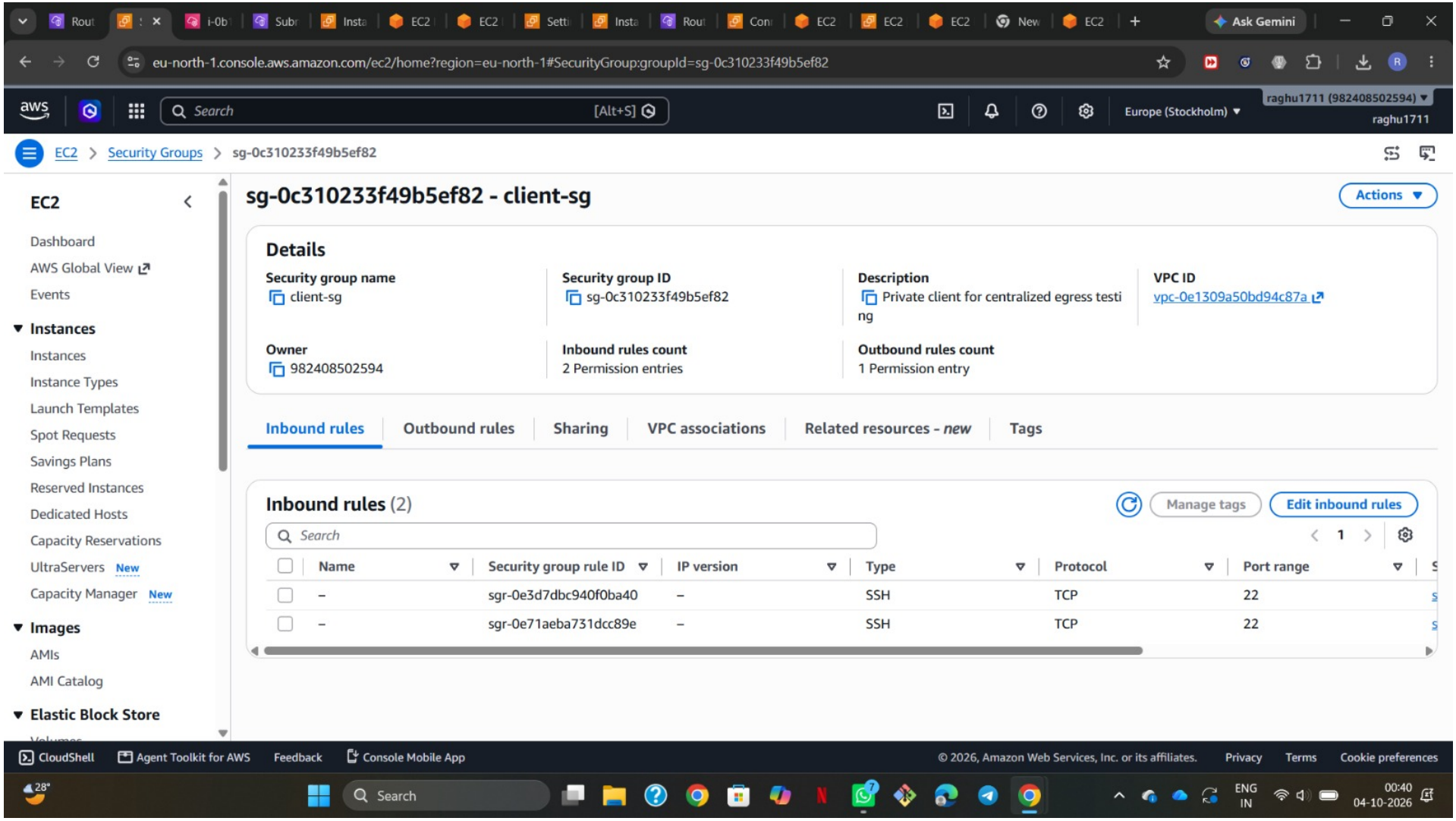
Shows the security-group configuration associated with the private egress client.



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Figure 11. Firewall Security Group

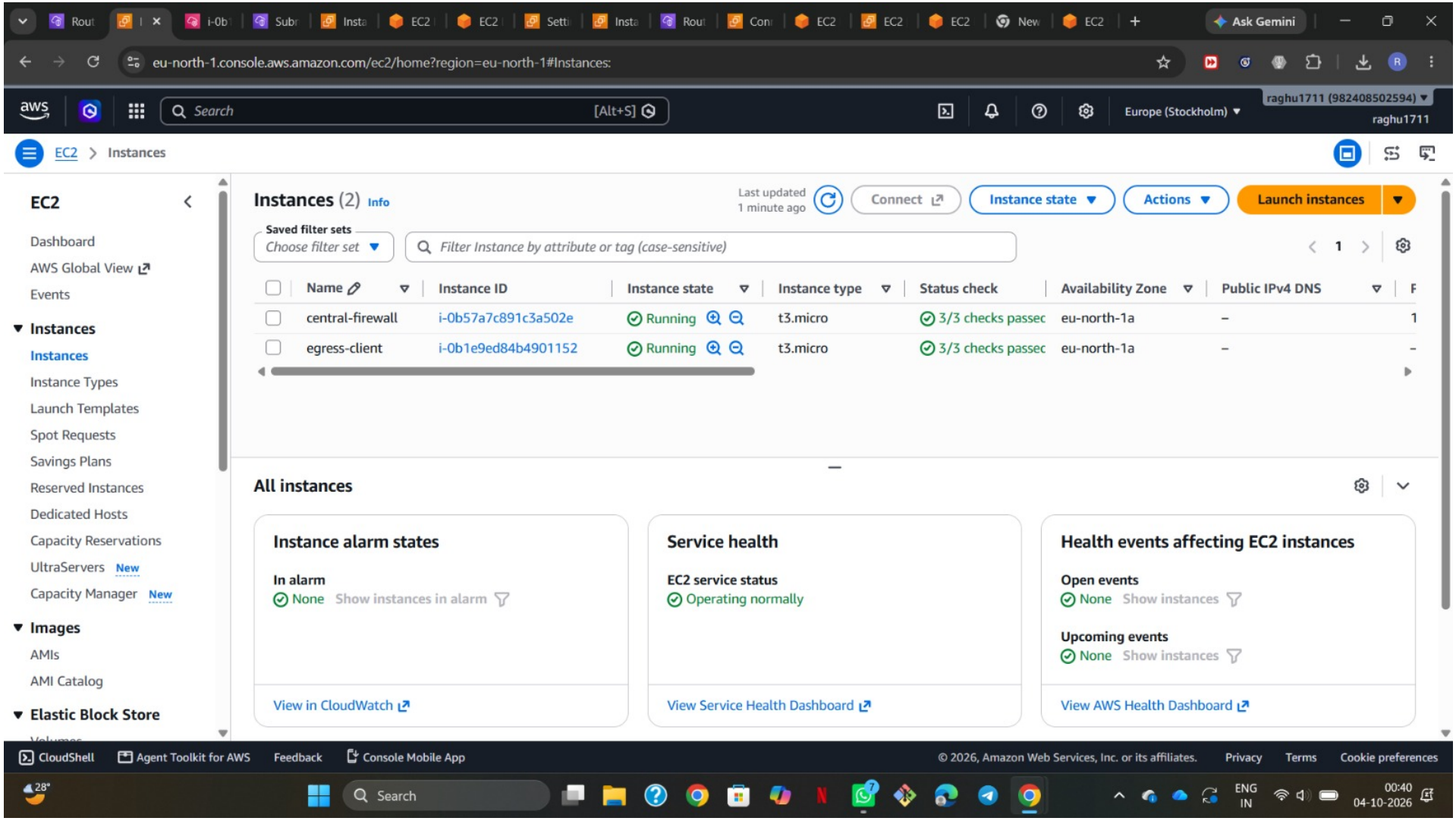
Shows the security-group configuration associated with the centralized firewall instance.



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Figure 12. EC2 Instances Overview

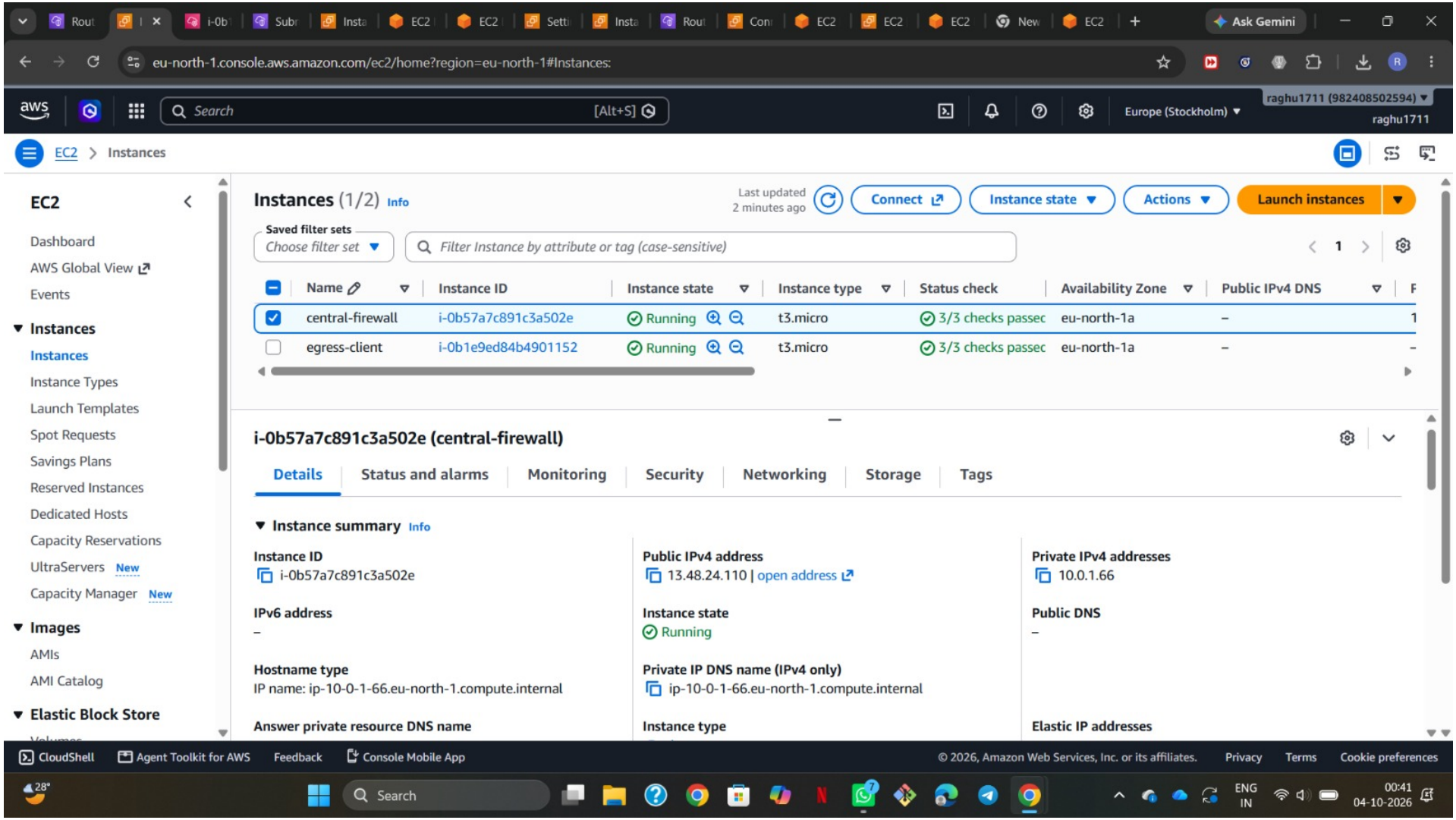
Shows the two EC2 instances used in the proof of concept: central-firewall and egress-client.



Source screenshot: WhatsApp Image 2026-10-04 at 12.40.47 AM.jpeg

Figure 13. Egress Client Instance

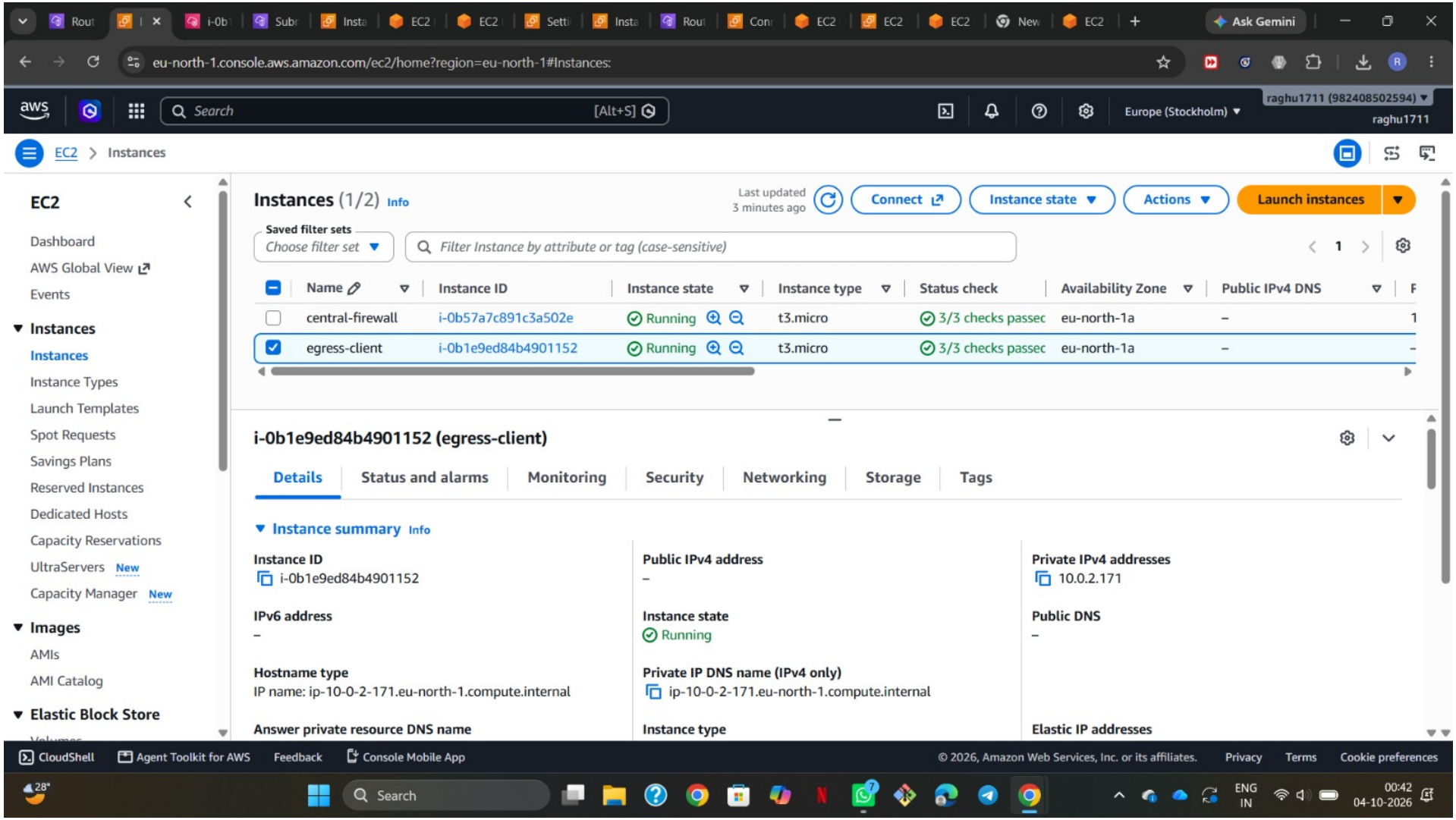
Shows the private client EC2 instance and its networking/instance details.



Source screenshot: WhatsApp Image 2026-10-04 at 12.41.40 AM.jpeg

Figure 14. Central Firewall Instance

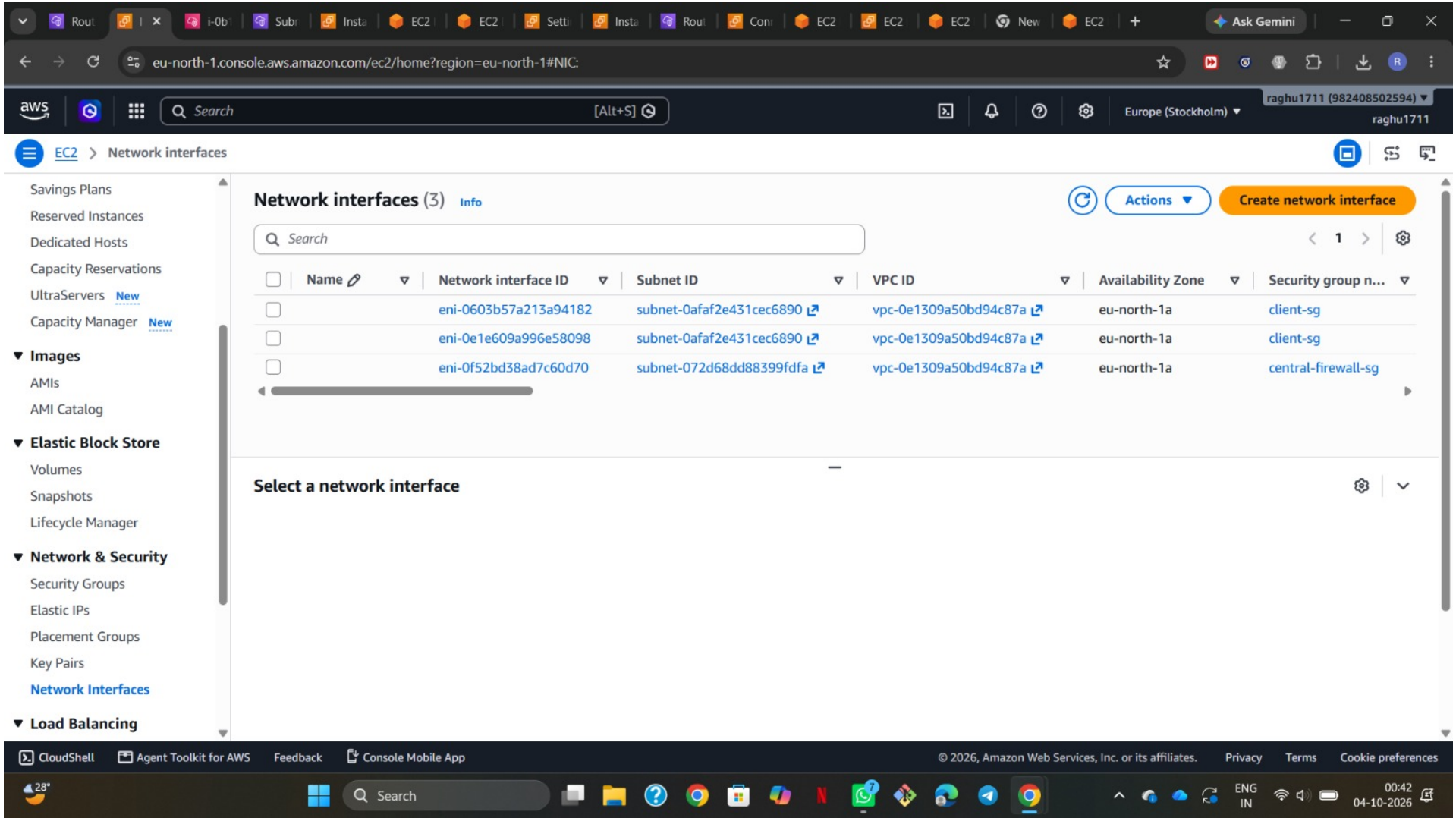
Shows the EC2 firewall/router instance and its networking/instance details.



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Figure 15. Firewall Network Interface

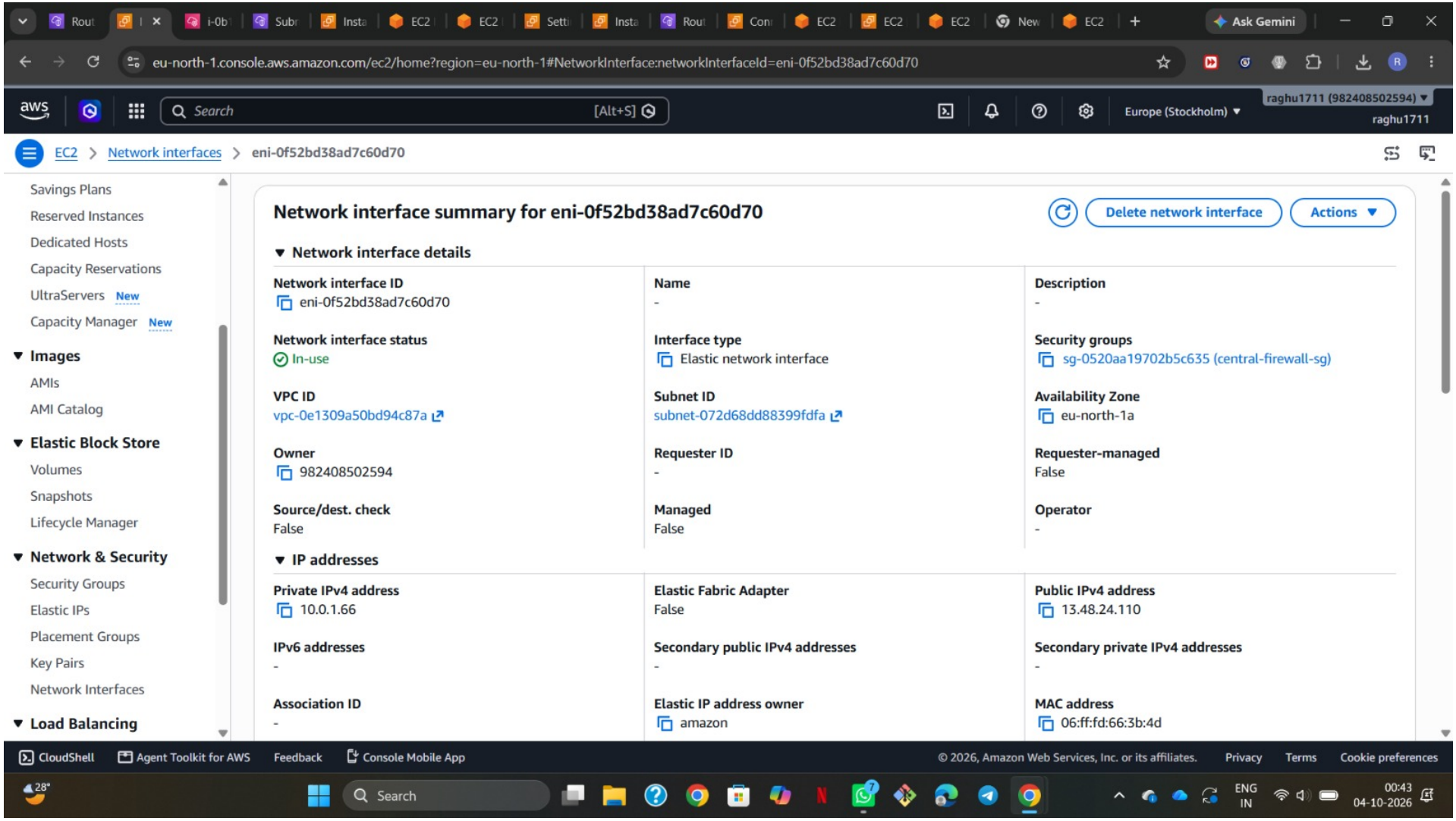
Shows the firewall ENI used as the next hop for private-subnet Internet traffic.



Source screenshot: WhatsApp Image 2026-10-04 at 12.42.54 AM.jpeg

Figure 16. Source/Destination Check Configuration

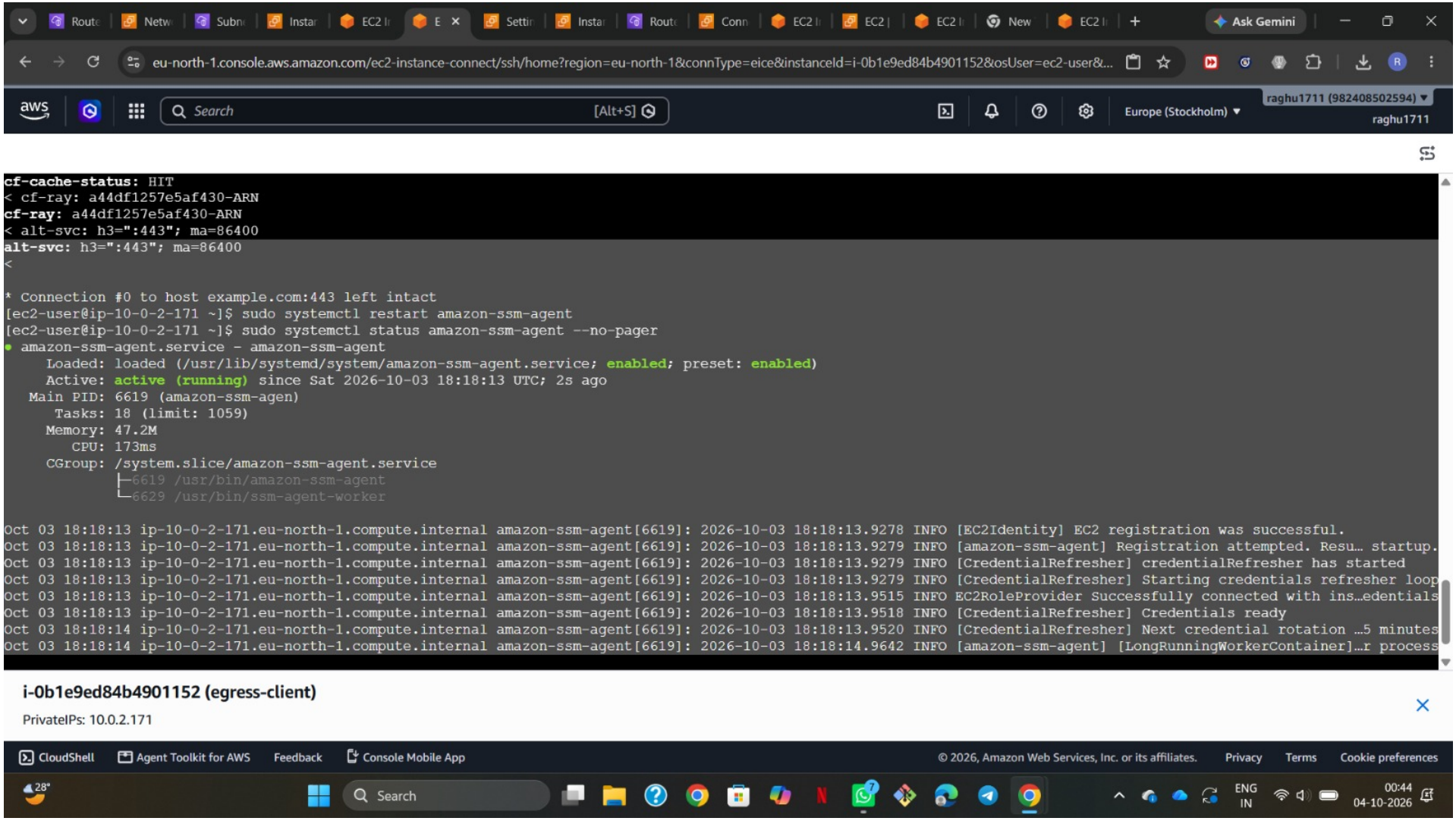
Shows the EC2 network-interface source/destination-check setting. This must be disabled when an EC2 instance forwards traffic for other hosts.



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Figure 17. Firewall Host Networking

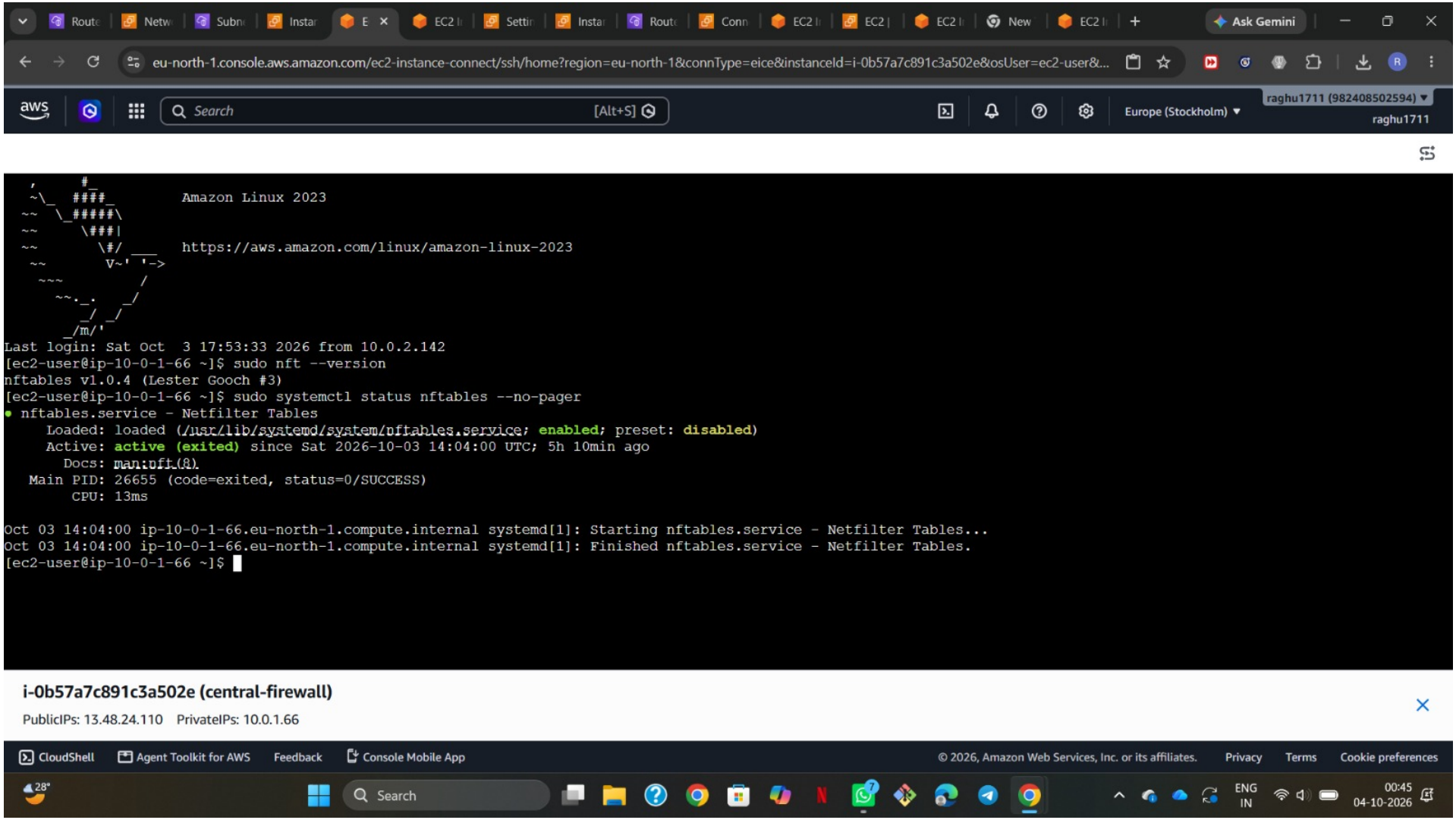
Terminal evidence from the firewall host showing Linux networking/firewall configuration used for packet forwarding.



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Figure 18. Client Operating-System Route

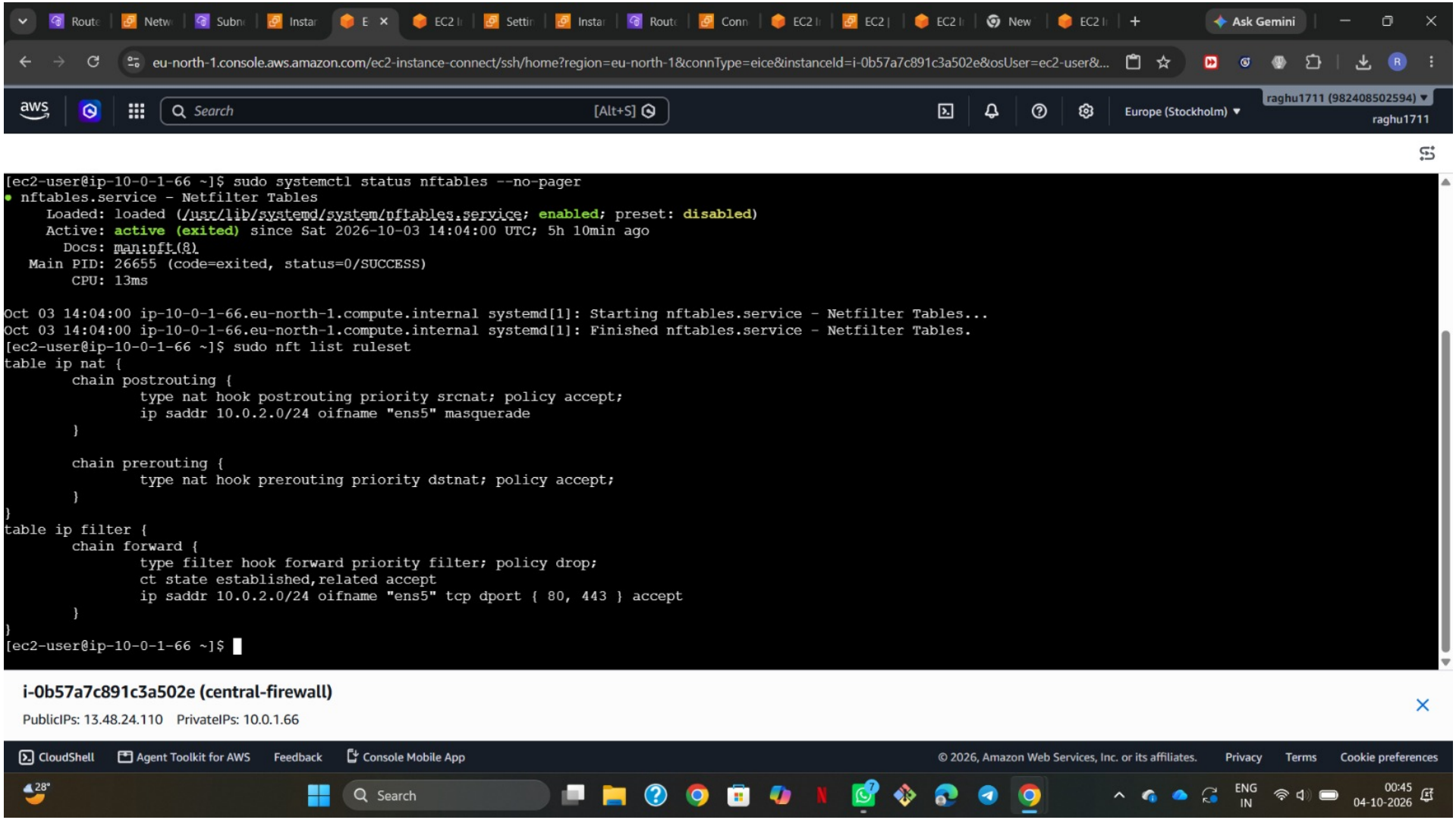
The client OS uses its subnet router (10.0.2.1) as the default gateway. AWS VPC routing then applies the subnet route table.



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Figure 19. Client Route Lookup

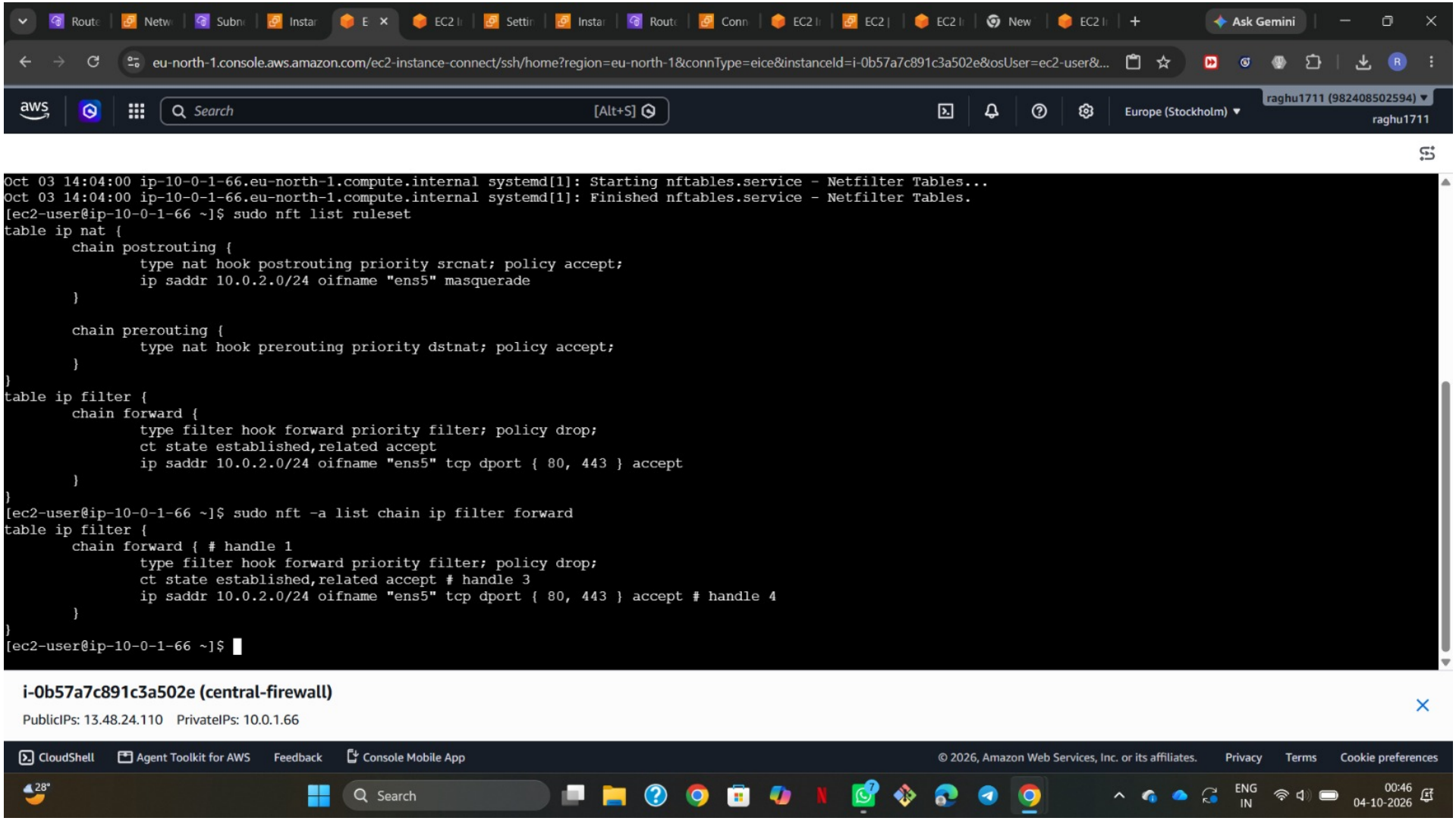
Shows a route lookup from the client for an external IPv4 address, confirming traffic leaves through the subnet gateway.



Source screenshot: WhatsApp Image 2026-10-04 at 12.45.39 AM.jpeg

Figure 20. Client-to-Firewall Reachability

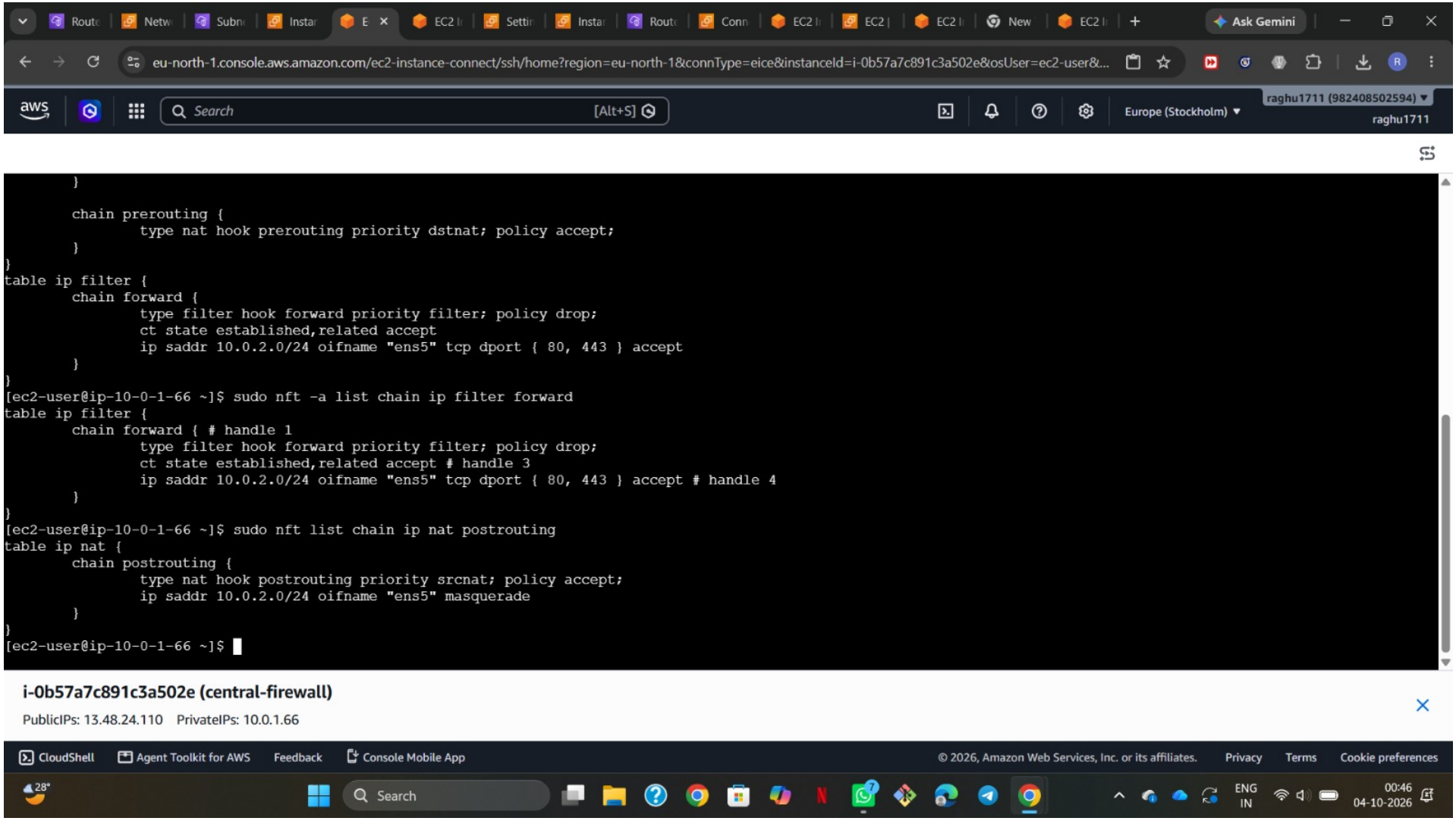
A connection to 10.0.1.66:443 is immediately refused rather than timing out. This demonstrates network reachability to the firewall host even though no service is listening on that port.



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Figure 21. HTTP Internet Test

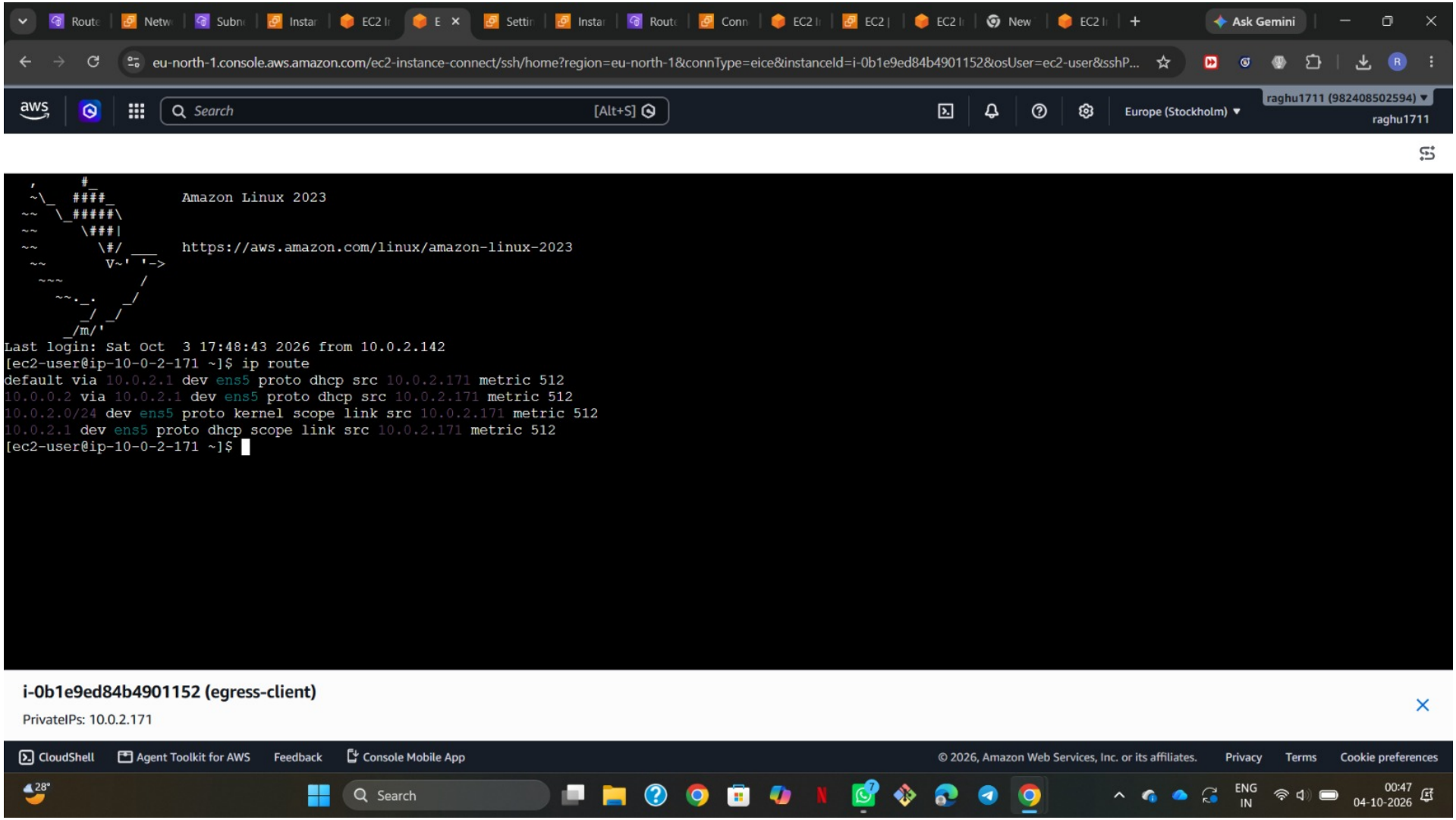
Successful HTTP response from example.com demonstrates that the private client can reach the Internet through the centralized egress path.



Source screenshot: WhatsApp Image 2026-10-04 at 12.46.34 AM.jpeg

Figure 22. Client Route Verification

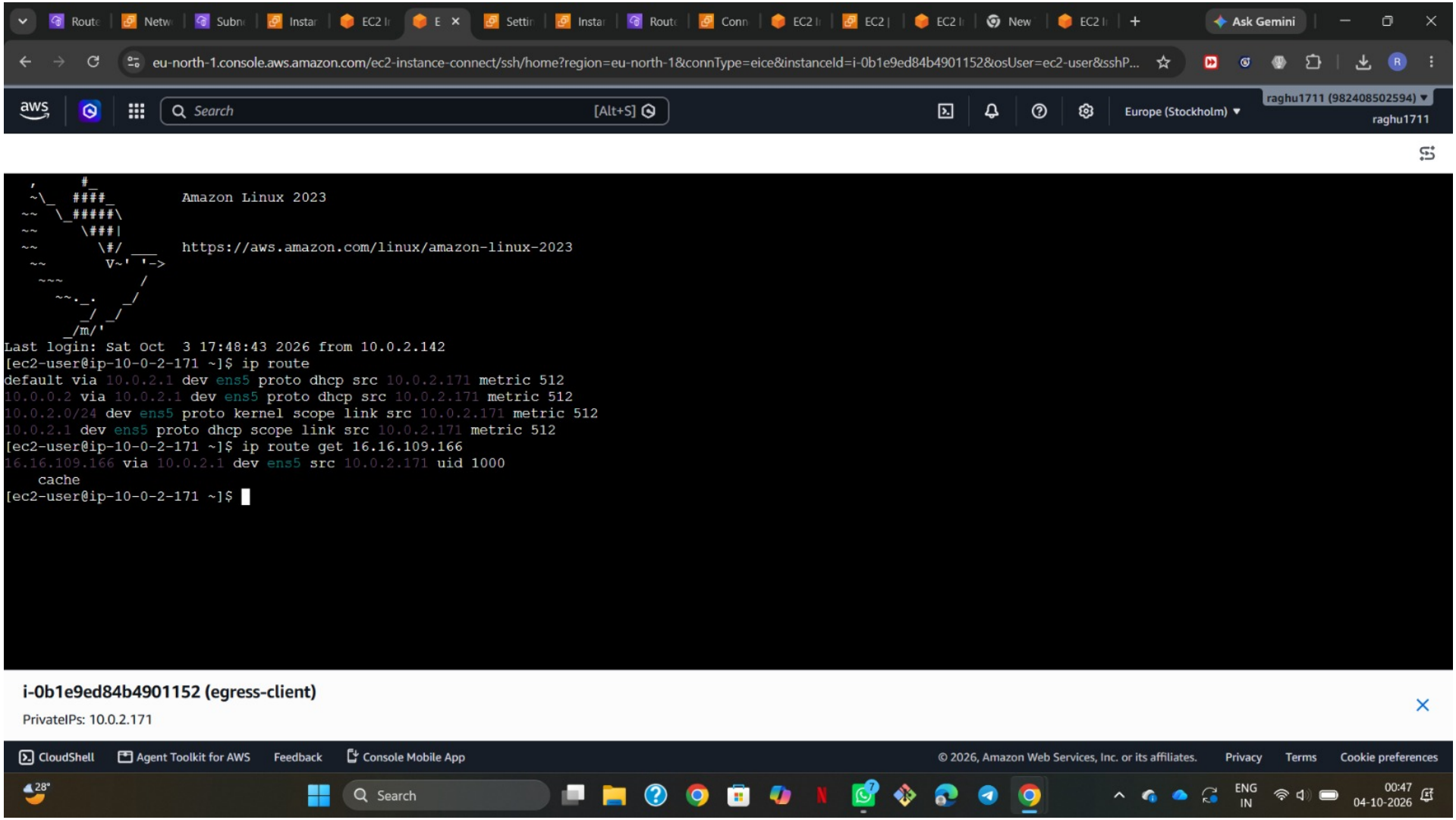
Additional client-side route evidence for the private EC2 instance.



Source screenshot: WhatsApp Image 2026-10-04 at 12.47.18 AM.jpeg

Figure 23. External Route Lookup

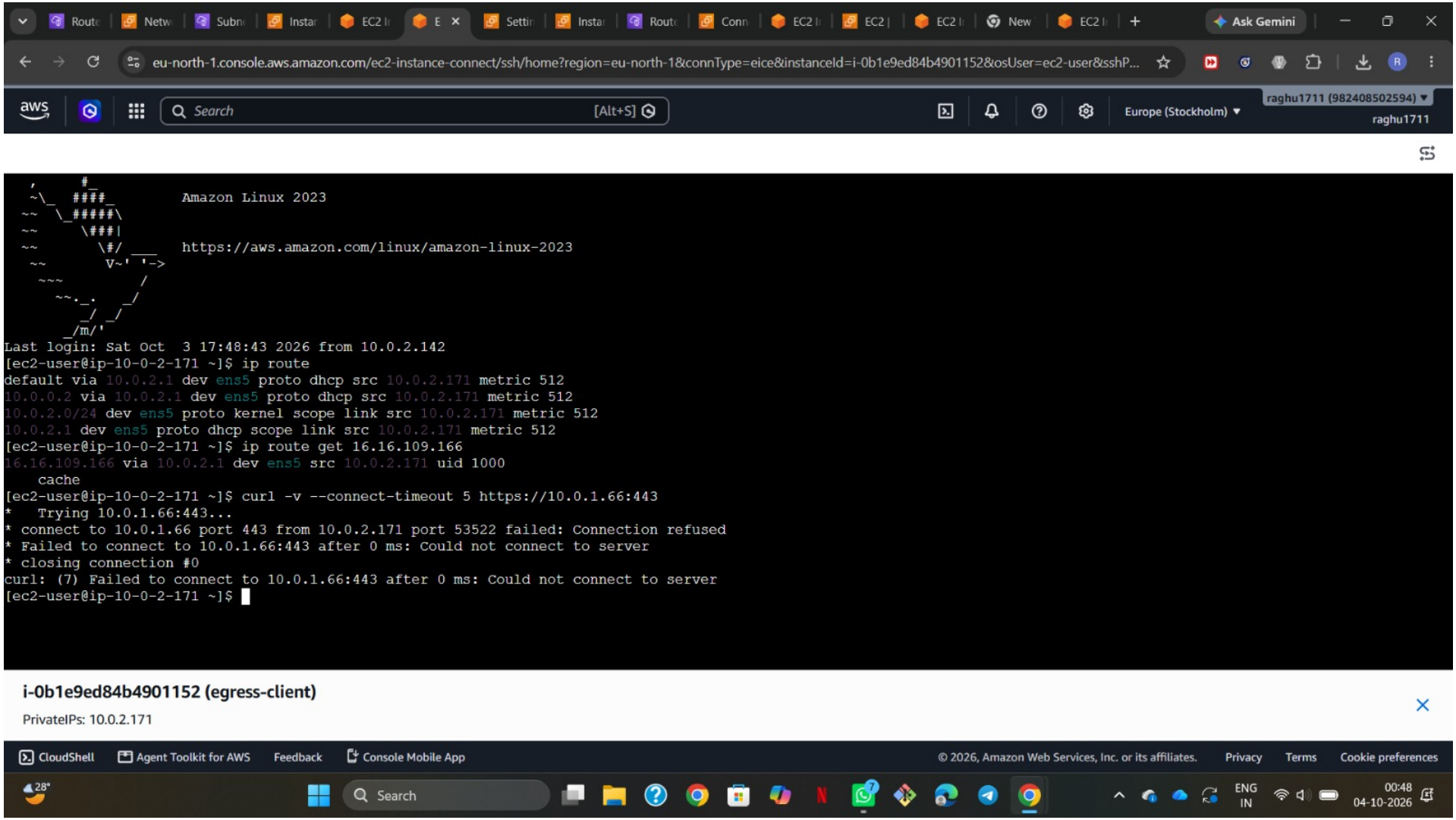
Additional route lookup evidence showing the client source address and subnet gateway used for external traffic.



Source screenshot: WhatsApp Image 2026-10-04 at 12.47.41 AM.jpeg

Figure 24. Firewall Reachability Test

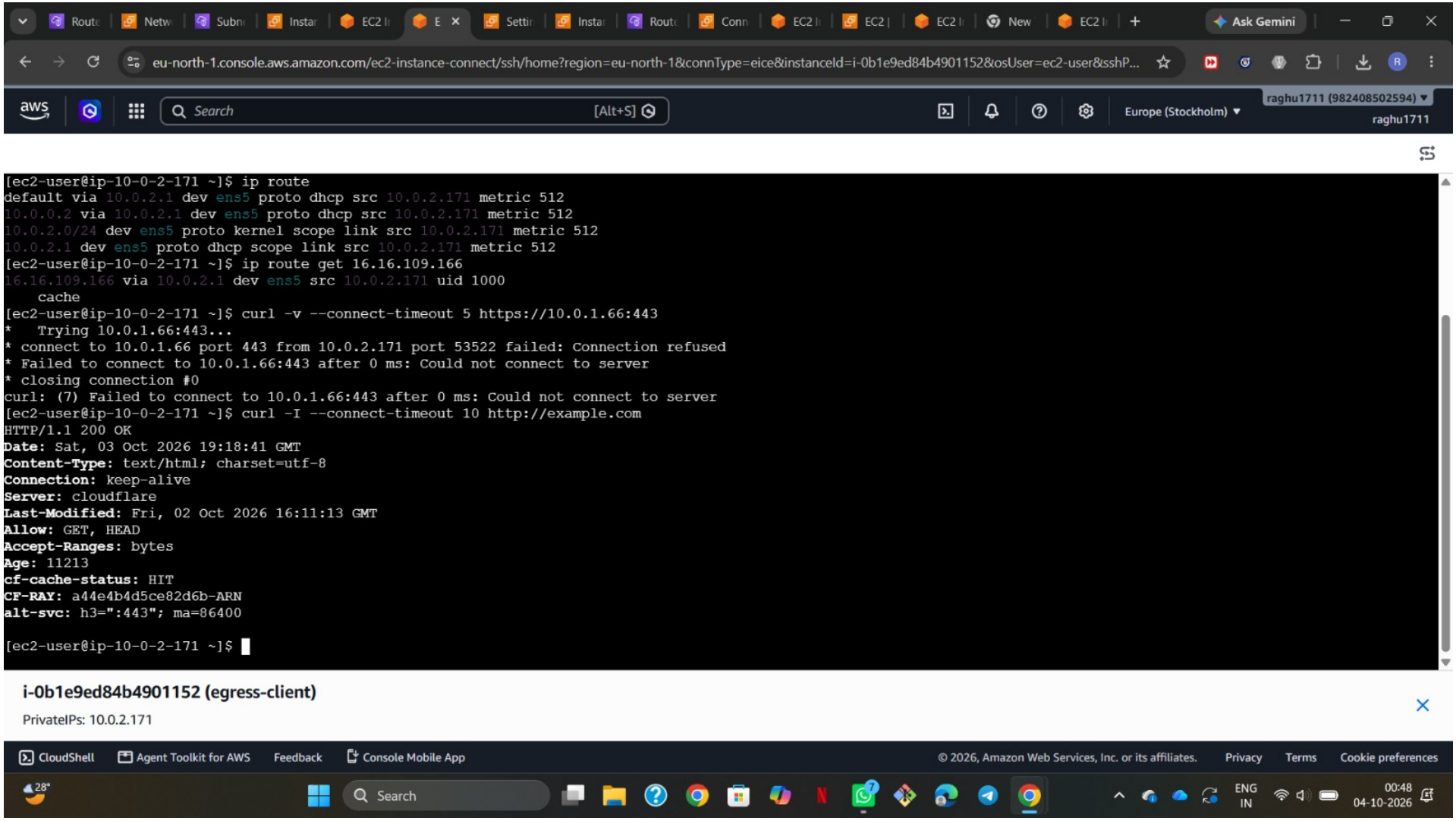
Additional client-to-firewall connectivity evidence using the firewall private address.



Source screenshot: WhatsApp Image 2026-10-04 at 12.48.26 AM.jpeg

Figure 25. HTTP Egress Success

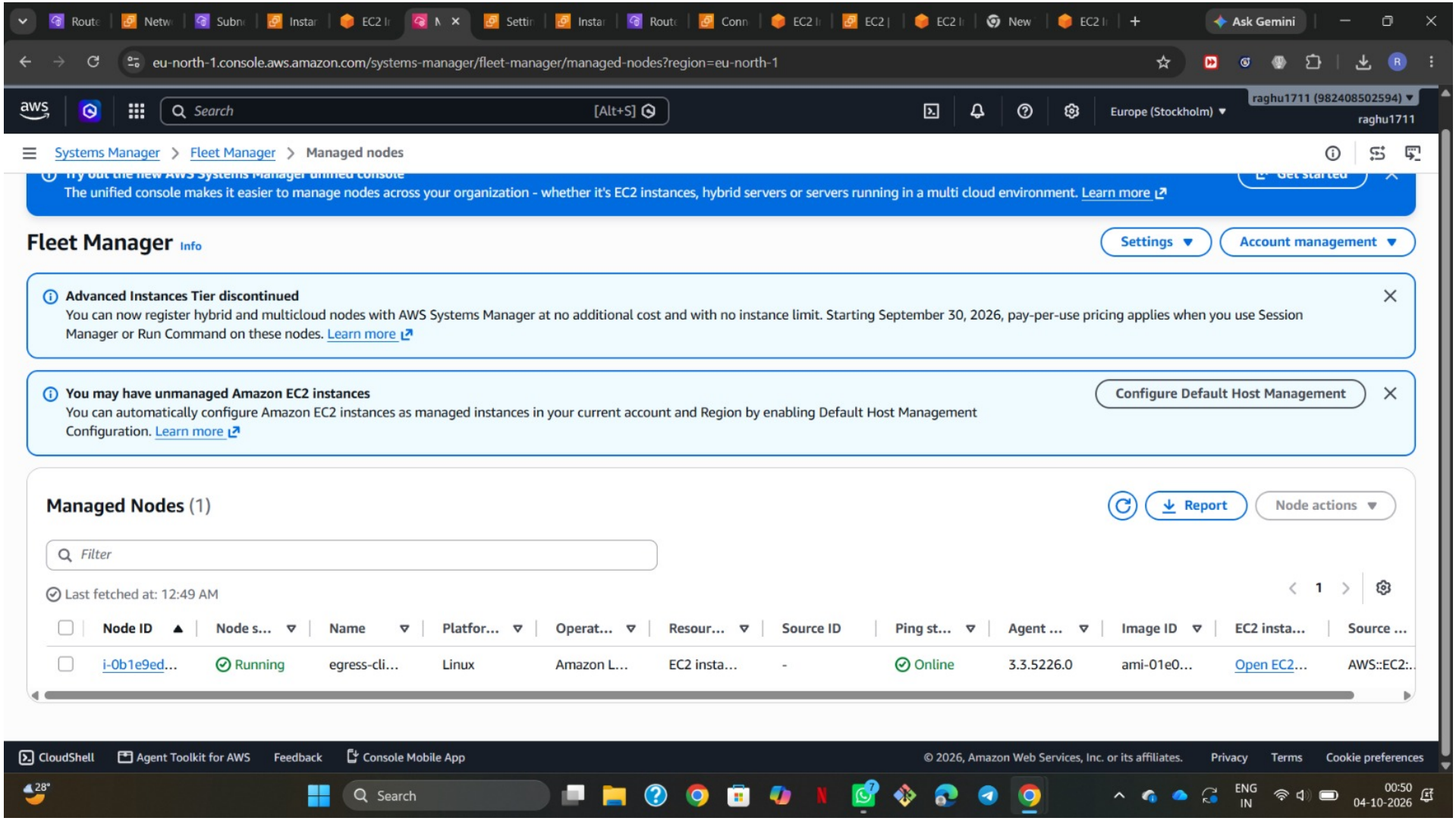
Successful HTTP/1.1 200 response provides clear proof that outbound web traffic from the private client is working.



Source screenshot: WhatsApp Image 2026-10-04 at 12.48.55 AM.jpeg

Figure 26. Systems Manager - Managed Node Online

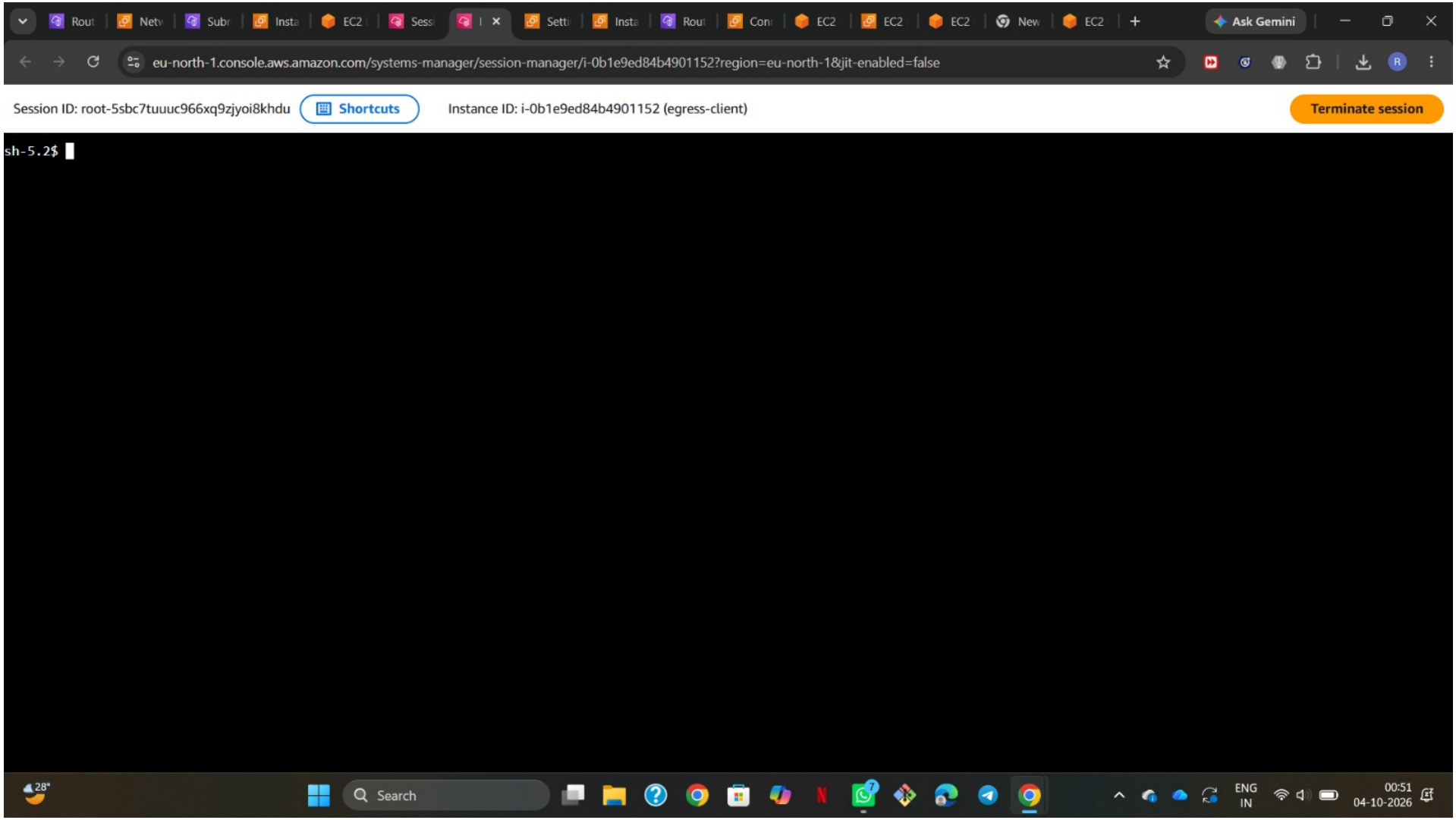
AWS Systems Manager Fleet Manager shows the egress-client as Online, demonstrating that outbound HTTPS connectivity is functioning.



Source screenshot: WhatsApp Image 2026-10-04 at 12.50.12 AM.jpeg

Figure 27. Systems Manager Session

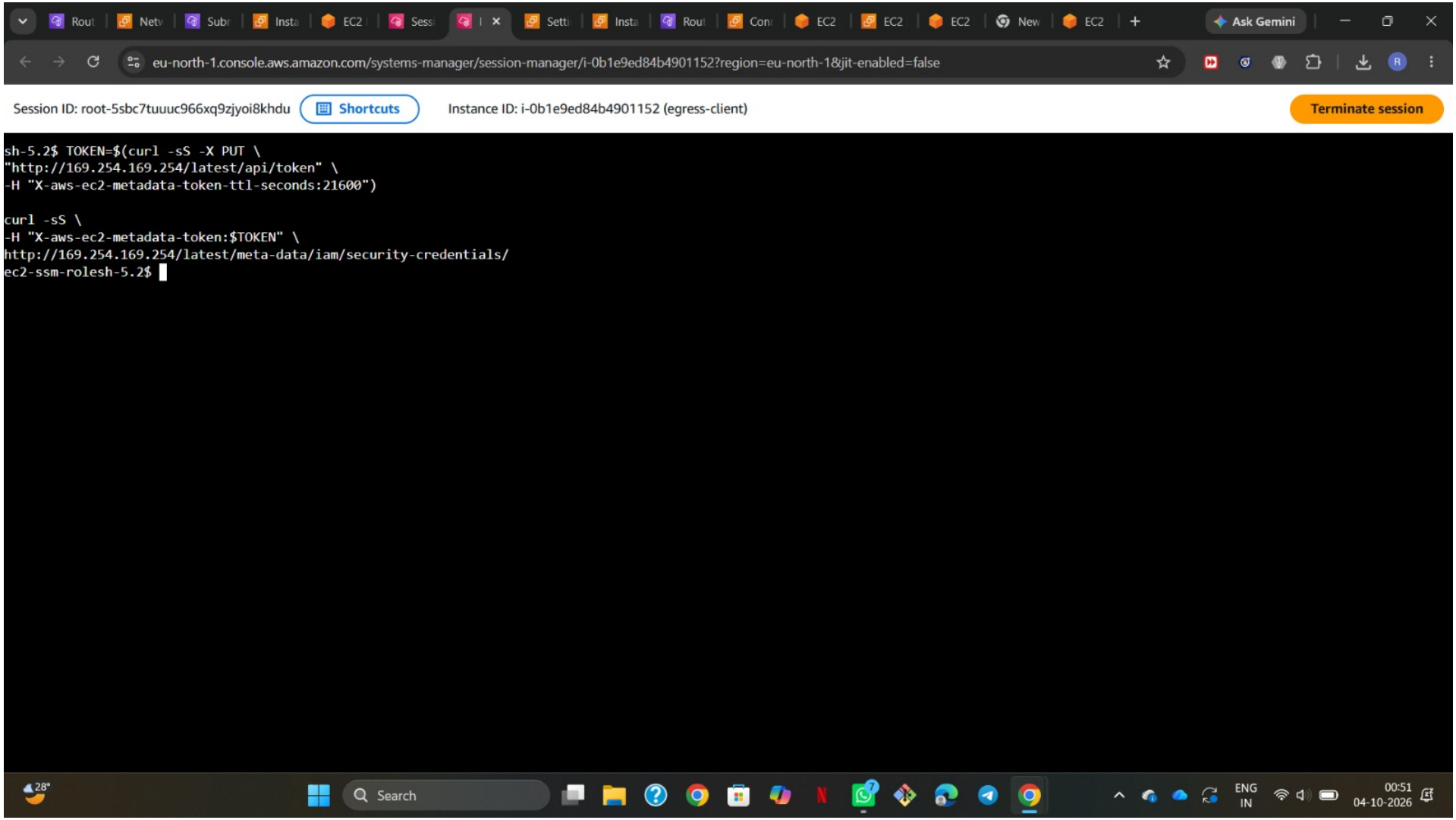
A live Session Manager terminal to the private egress-client demonstrates management access without assigning the client a public IPv4 address.



Source screenshot: WhatsApp Image 2026-10-04 at 12.51.05 AM.jpeg

Figure 28. IAM Role Verification from IMDSv2

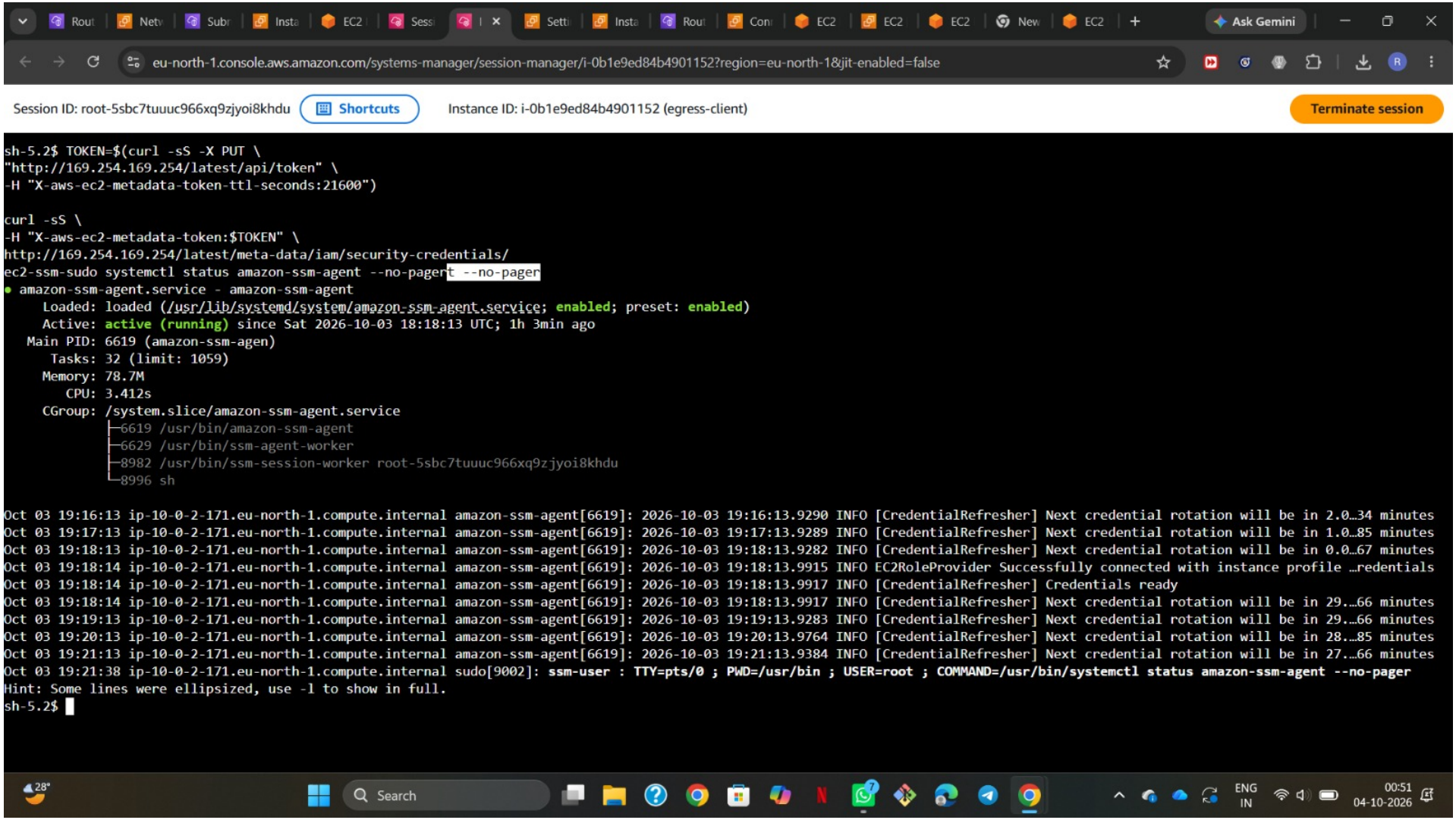
The instance metadata service returns the attached role name ec2-ssm-role, confirming the client has the IAM role required for Systems Manager.



Source screenshot: WhatsApp Image 2026-10-04 at 12.51.29 AM.jpeg

Figure 29. SSM Agent Status

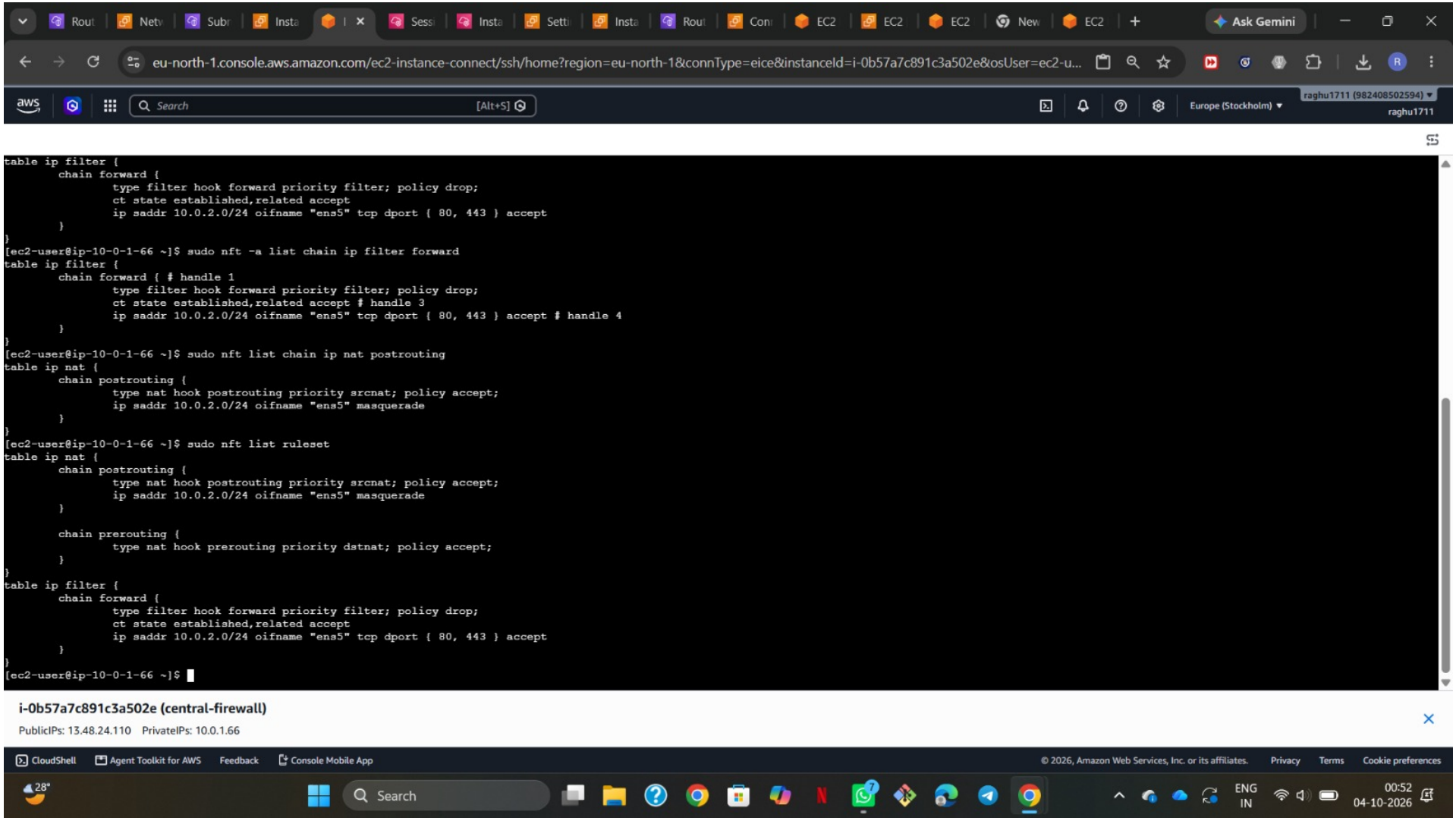
The amazon-ssm-agent service is active and running, and the log output shows successful instance-profile credential access.



Source screenshot: WhatsApp Image 2026-10-04 at 12.51.52 AM.jpeg

Figure 30. Final nftables Ruleset

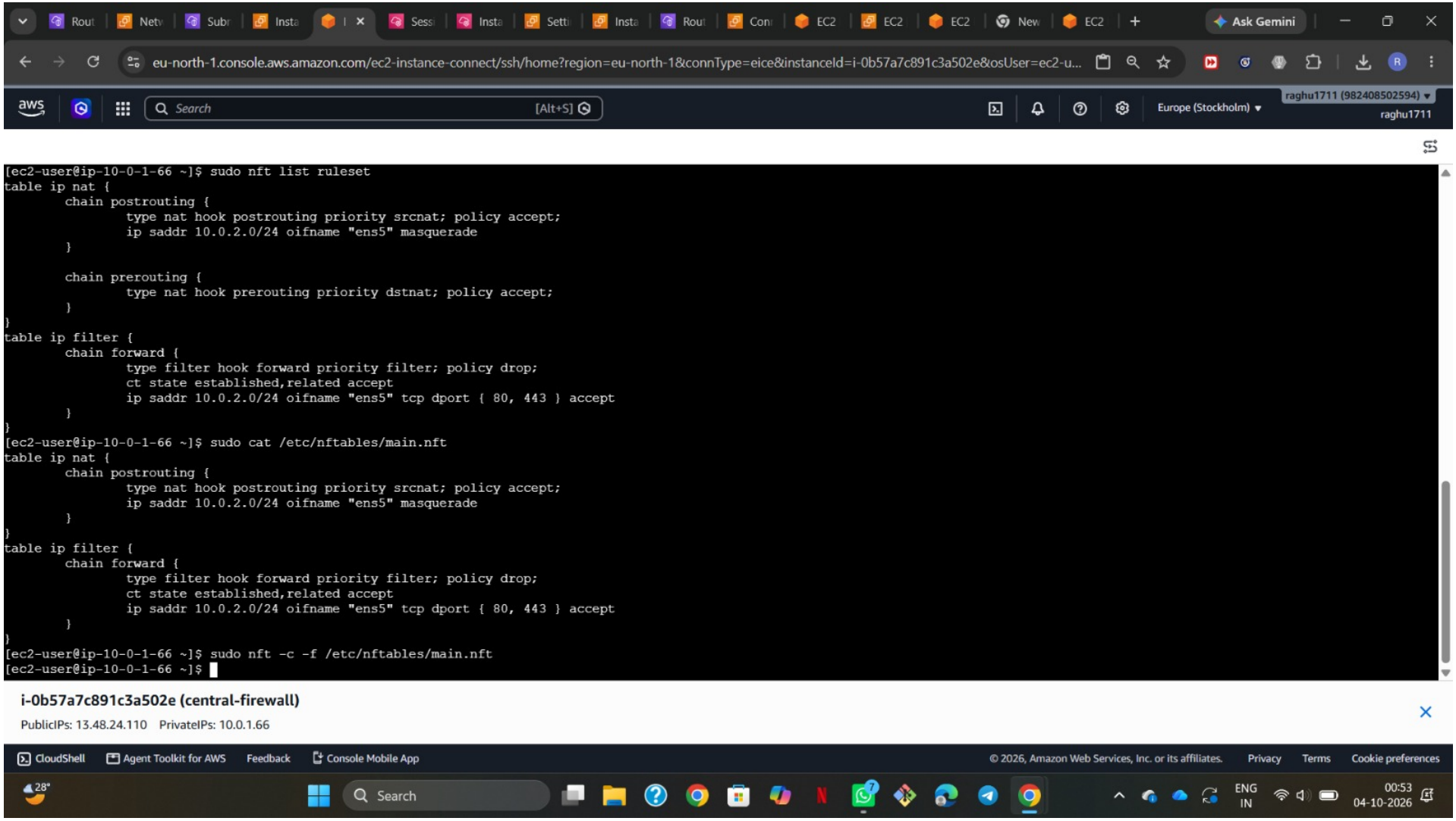
Shows the final centralized firewall rules: stateful return traffic, HTTP/HTTPS forwarding for 10.0.2.0/24, default-drop forwarding policy, and source NAT masquerading.



Source screenshot: WhatsApp Image 2026-10-04 at 12.52.30 AM.jpeg

Figure 31. Firewall Rules Persistence Check

Shows the saved /etc/nftables/main.nft configuration and a successful nft syntax check, providing evidence that the final rules are valid and prepared for persistence.



Source screenshot: WhatsApp Image 2026-10-04 at 12.53.13 AM.jpeg

4. Project Result

The collected evidence demonstrates a working centralized egress-filtering proof of concept. The egress-client is placed in a private subnet without a public IPv4 address. Its Internet-bound traffic is routed through the central-firewall EC2 instance. The firewall uses nftables to apply a default-drop forwarding policy, allows outbound HTTP/HTTPS traffic from the client subnet, and uses masquerading for Internet access. AWS Systems Manager subsequently reports the client Online and provides a working Session Manager connection.

Scope note: This is an EC2-based firewall/router proof of concept designed to demonstrate centralized egress control while avoiding the higher-cost managed architecture components. It should not be presented as an AWS Network Firewall deployment.

Key troubleshooting lesson: the decisive networking fix was disabling source/destination checking on the firewall ENI so that the EC2 instance could legally forward traffic for the private client.